FULL MOLECULAR HAMILTONIAN

From Classical to Quantum

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Contents

1 Classical Cartesian Hamiltonian

We have N_{nucl} nuclei labeled by j with masses M_j, charges Z_je, positions $\mathbf{R}_j$, and conjugate momenta $\mathbf{P}_j$; and N_{elec} electrons labeled by i with mass m_e, charge $-e$, positions $\mathbf{r}_i$, and momenta $\mathbf{p}_i$. All vectors live in a space-fixed (SF) lab frame. Using SI units and writing $R_{ij} = |\mathbf{R}_i - \mathbf{R}_j|$ etc.,

$$H = \underbrace{\sum_{j=1}^{N_{\mathrm{nucl}}} \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_{i=1}^{N_{\mathrm{elec}}} \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \tag{1.1}$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{i\neq j}\frac{Z_iZ_je^2}{R_{ij}} - \sum_{i,j}\frac{Z_je^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}}\right). \tag{1.2}$$

Canonical Poisson brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk}\delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij}\delta_{\alpha\beta}$.

2 Direct Transformation to Mass-Centered Coordinates

The translational coordinate and the internal reference point need not be the same, and choosing them *differently* is what makes the transformation land directly on the final form: take the *total* (nuclei + electrons) center of mass as the translational coordinate — so the free term carries the correct total mass — but refer all *internal* coordinates, nuclear and electronic, to the *nuclear* center of mass — the Born–Oppenheimer-friendly reference. A single point transformation then produces the exact CoM separation *and* the electronic mass-polarization term at once, with no subsequent canonical transformation.

2.1 New coordinates

Define the nuclear and the total centers of mass,

$$\mathbf{R}_{\mathrm{ncm}} = \frac{1}{M_N}\sum_j M_j\mathbf{R}_j, \qquad M_N = \sum_j M_j, \tag{2.1}$$

$$\mathbf{R}_{\mathrm{cm}} = \frac{1}{M_{\mathrm{tot}}}\Big(\sum_j M_j\mathbf{R}_j + m_e\sum_i \mathbf{r}_i\Big) = \mathbf{R}_{\mathrm{ncm}} + \frac{m_e}{M_{\mathrm{tot}}}\sum_i\big(\mathbf{r}_i - \mathbf{R}_{\mathrm{ncm}}\big), \tag{2.2}$$

with $M_{\mathrm{tot}} = M_N + N_{\mathrm{elec}}\,m_e$, and take as new coordinates the total CoM $\mathbf{R}_{\mathrm{cm}}$ together with internal coordinates referred to the *nuclear* CoM:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\mathrm{ncm}}, \qquad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\mathrm{ncm}}. \tag{2.3}$$

Only the nuclear relative vectors are linearly dependent:

$$\sum_j M_j\mathbf{R}'_j = 0, \tag{2.4}$$

the electron coordinates $\mathbf{r}'_i$ being unconstrained. The second equality in (2.2) shows how the two reference points sit relative to one another: the total CoM is the nuclear CoM displaced by the (mass-weighted) electronic polarization $\frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}'_i$.

2.2 Canonical momenta

This is a point transformation $\mathbf{Q}'_{\text{new}} = \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$ with $\mathbf{q}_{\text{old}} = (\mathbf{R}_j, \mathbf{r}_i)$ and $\mathbf{Q}'_{\text{new}} = (\mathbf{R}_{\text{cm}}, \mathbf{R}'_j, \mathbf{r}'_i)$. Use the generating function: $F_2 = F_2(\mathbf{q}_{\text{old}}, \mathbf{P}, t) = \mathbf{P} \cdot \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$

$$F_2 = \mathbf{P}_{\text{cm}} \cdot \mathbf{R}_{\text{cm}}(\mathbf{R}_j, \mathbf{r}_i) + \sum_j \mathbf{P}'_j \cdot \mathbf{R}'_j(\{\mathbf{R}_j\}) + \sum_i \mathbf{p}'_i \cdot \mathbf{r}'_i(\mathbf{R}_j, \mathbf{r}_i). \tag{2.5}$$

Then $\mathbf{P}_j = \partial F_2/\partial \mathbf{R}_j$, $\mathbf{p}_i = \partial F_2/\partial \mathbf{r}_i$. Differentiating (2.1)–(2.3) with respect to a nuclear position,

$$\frac{\partial \mathbf{R}_{\text{cm}}}{\partial \mathbf{R}_j} = \frac{M_j}{M_{\text{tot}}} \mathbb{1}, \qquad \frac{\partial \mathbf{R}'_k}{\partial \mathbf{R}_j} = \Big(\delta_{jk} - \frac{M_j}{M_N}\Big) \mathbb{1}, \qquad \frac{\partial \mathbf{r}'_l}{\partial \mathbf{R}_j} = -\frac{M_j}{M_N} \mathbb{1}, \tag{2.6a}$$

and with respect to an electron position,

$$\frac{\partial \mathbf{R}_{\text{cm}}}{\partial \mathbf{r}_i} = \frac{m_e}{M_{\text{tot}}} \mathbb{1}, \qquad \frac{\partial \mathbf{R}'_k}{\partial \mathbf{r}_i} = 0, \qquad \frac{\partial \mathbf{r}'_l}{\partial \mathbf{r}_i} = \delta_{il}\, \mathbb{1}. \tag{2.6b}$$

The mixed choice is visible here: the translational row carries the *total*-mass fractions M_j/M_{tot}, m_e/M_{tot}, while the internal rows carry the *nuclear* fractions M_j/M_N — and the electron positions do not move the internal reference point at all, $\partial \mathbf{R}'_k/\partial \mathbf{r}_i = 0$. One obtains

$$\mathbf{P}_j = \frac{M_j}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} + \mathbf{P}'_j - \frac{M_j}{M_N} \Big(\textstyle\sum_k \mathbf{P}'_k + \sum_l \mathbf{p}'_l \Big), \qquad \mathbf{p}_i = \tfrac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} + \mathbf{p}'_i. \tag{2.7}$$

The constraint dual to (2.4) is $\sum_k \mathbf{P}'_k = 0$, so

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} + \mathbf{P}'_j - \frac{M_j}{M_N} \sum_l \mathbf{p}'_l, \qquad \mathbf{p}_i = \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} + \mathbf{p}'_i.} \tag{2.8}$$

Each particle carries its total-mass-fraction share of $\mathbf{P}_{\text{cm}}$ plus its internal momentum; in addition, each *nucleus* carries the recoil $-\frac{M_j}{M_N} \sum_l \mathbf{p}'_l$ against the electrons, because the internal reference point (2.1) is built from the nuclei alone. Summing (2.8) over all particles, the mass fractions add to one, $\sum_j \frac{M_j}{M_N} = 1$ makes the recoils cancel against $\sum_i \mathbf{p}'_i$, and $\sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i = \mathbf{P}_{\text{cm}}$: the new translational momentum is the total momentum, as it must be.

2.3 Hamiltonian after the transformation

Insert (2.8) into T_N in (1.1). The nuclear momentum has the form $\mathbf{P}_j = M_j \mathbf{b} + \mathbf{P}'_j$ with the common vector

$$\mathbf{b} \equiv \frac{\mathbf{P}_{\text{cm}}}{M_{\text{tot}}} - \frac{1}{M_N} \sum_l \mathbf{p}'_l, \tag{2.9}$$

so, expanding the square and using the dual constraint $\sum_j \mathbf{P}'_j = 0$ on the cross term,

$$T_N = \sum_j \frac{(M_j \mathbf{b} + \mathbf{P}'_j)^2}{2M_j} = \frac{M_N}{2}\,\mathbf{b}^2 + \mathbf{b}\cdot \underbrace{\sum_j \mathbf{P}'_j}_{=0} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j}, \tag{2.10a}$$

where, squaring (2.9),

$$\frac{M_N}{2}\,\mathbf{b}^2 = \frac{M_N}{M_{\rm tot}}\frac{\mathbf{P}^2_{\rm cm}}{2M_{\rm tot}} - \frac{\mathbf{P}_{\rm cm}\cdot\sum_l \mathbf{p}'_l}{M_{\rm tot}} + \frac{\left(\sum_l \mathbf{p}'_l\right)^2}{2M_N}. \tag{2.10b}$$

The electrons give

$$T_e = \sum_i \frac{1}{2m_e}\left[\frac{m_e}{M_{\rm tot}}\mathbf{P}_{\rm cm} + \mathbf{p}'_i\right]^2 = \frac{N_{\rm elec} m_e}{M_{\rm tot}}\frac{\mathbf{P}^2_{\rm cm}}{2M_{\rm tot}} + \frac{\mathbf{P}_{\rm cm}\cdot\sum_i \mathbf{p}'_i}{M_{\rm tot}} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e}. \tag{2.11}$$

Adding (2.10a)–(2.10b) and (2.11): the two mass fractions in front of $\mathbf{P}^2_{\rm cm}/2M_{\rm tot}$ sum to $(M_N + N_{\rm elec} m_e)/M_{\rm tot} = 1$, and the two cross terms cancel *identically* — the nuclear recoil against the electrons contributes $-\mathbf{P}_{\rm cm}\cdot\sum \mathbf{p}'/M_{\rm tot}$, and the electrons' share of the CoM momentum contributes exactly $+\mathbf{P}_{\rm cm}\cdot\sum \mathbf{p}'/M_{\rm tot}$. No constraint is needed for this cancellation; it is built into the coordinate choice. What survives of $\frac{M_N}{2}\mathbf{b}^2$ is the **mass-polarization** term. Combining with V:

$$\boxed{H = \frac{\mathbf{P}^2_{\rm cm}}{2M_{\rm tot}} + \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e} + \underbrace{\frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2}_{\text{electron mass polarization}} + V(\mathbf{R}'_j, \mathbf{r}'_i).} \tag{2.12}$$

The potential is unchanged in form because both internal families are referred to the *same* point $\mathbf{R}_{\rm ncm}$: $\mathbf{R}'_j - \mathbf{R}'_k = \mathbf{R}_j - \mathbf{R}_k$, $\mathbf{R}'_j - \mathbf{r}'_i = \mathbf{R}_j - \mathbf{r}_i$, $\mathbf{r}'_i - \mathbf{r}'_l = \mathbf{r}_i - \mathbf{r}_l$. Because H is independent of $\mathbf{R}_{\rm cm}$, $\mathbf{P}_{\rm cm}$ is conserved and the CoM separates exactly, *for any* $\mathbf{P}_{\rm cm}$: the free term carries the correct total mass $M_{\rm tot}$, there is no COM–internal cross-coupling, and the internal Hamiltonian

$$\boxed{H_{\rm int} = \sum_j \frac{\mathbf{P}'^2_j}{2M_j} + \sum_i \frac{\mathbf{p}'^2_i}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2 + V(\mathbf{R}'_j, \mathbf{r}'_i)} \tag{2.13}$$

already contains the mass polarization, $(M_N)^{-1}\sum_{i<l} \mathbf{p}'_i\cdot\mathbf{p}'_l + (2M_N)^{-1}\sum_i \mathbf{p}'^2_i$, produced by the coordinate choice itself — no subsequent canonical (Galilean-boost) transformation is required.

2.4 Remark: anatomy of the coordinate choice

Each term in (2.12) can be traced to one design decision:

- *Translational coordinate = total CoM* puts the total-mass fractions $M_j/M_{\rm tot}$, $m_e/M_{\rm tot}$ into the momentum relations (2.8), which is what makes the $\mathbf{P}^2_{\rm cm}$ coefficients sum to $1/2M_{\rm tot}$ and the cross terms cancel identically.

- *Electrons referred to the nuclear CoM* gives the nuclei the recoil $-\frac{M_j}{M_N}\sum_l \mathbf{p}'_l$ in (2.8); squaring it in T_N is the sole source of the mass polarization $(\sum_i \mathbf{p}'_i)^2/2M_N$.

- *Nuclei referred to the nuclear CoM* keeps every interparticle separation a plain coordinate difference, so V is form-invariant, and keeps the constraint (2.4) purely nuclear.

The nuclear reference point is the one place where an alternative suggests itself: referring the nuclei to the *total* CoM instead, $\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\rm cm}$, leads (with the analogous gauge $\sum_k \mathbf{P}'_k = 0$) to the *same* momentum relations (2.8) and hence the same kinetic decomposition (2.12), mass polarization included. But the coordinate-side bookkeeping degrades: the constraint becomes the mixed relation $\sum_j M_j\mathbf{R}'_j + \frac{M_N m_e}{M_{\rm tot}}\sum_i \mathbf{r}'_i = 0$, and, because the two internal families no longer share a reference point, the electron–nucleus separations are no longer coordinate differences,

$$\mathbf{R}_j - \mathbf{r}_i \;=\; \mathbf{R}'_j - \mathbf{r}'_i + \frac{m_e}{M_{\rm tot}}\sum_l \mathbf{r}'_l, \tag{2.14}$$

so V_{Ne} picks up a polarization shift in its argument. Sharing the nuclear CoM between both internal families avoids both blemishes at no kinetic cost. Compared instead with referring *everything* to the total CoM, where the internal kinetic energy is diagonal but one constraint ties all particles together, the present choice trades that diagonality for the mass-polarization term while keeping the electrons unconstrained — the trade that suits the Born–Oppenheimer treatment. It is the internal Hamiltonian (2.13), exact for *any* $\mathbf{P}_{\rm cm}$, that we carry into the body-fixed treatment of Section 3.

3 BF frame for both nuclei and electrons

3.1 Setup

With Eckart's frame [1] fixed by the *nuclei*,

$$\mathbf{R}'_j = \mathbf{S}(\phi,\theta,\chi)\,\bar{\mathbf{R}}_j(Q), \qquad \mathbf{r}'_i = \mathbf{S}(\phi,\theta,\chi)\,\bar{\mathbf{r}}_i. \tag{3.1}$$

The Eckart conditions $\sum_j M_j\bar{\mathbf{R}}_j = 0$ and $\sum_j M_j\bar{\mathbf{R}}_j^{\rm eq}\times\bar{\mathbf{R}}_j = 0$ fix the orientation of $\mathbf{S}$ from the *nuclear* positions; the same $\mathbf{S}$ is then used to define BF electron coordinates $\bar{\mathbf{r}}_i$. The nuclear shape is parametrized by rectilinear mass-weighted **normal coordinates** Q_b,

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\rm eq} + \frac{1}{\sqrt{M_j}}\sum_b \mathbf{l}_{jb}\,Q_b, \tag{3.1a}$$

with mass-weighted normal-mode vectors $\mathbf{l}_{jb}$ obeying orthonormality and the Eckart (Sayvetz) conditions [1, 2]

$$\sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}, \qquad \sum_j \sqrt{M_j}\,\mathbf{l}_{jb} = 0, \qquad \sum_j \sqrt{M_j}\,\bar{\mathbf{R}}_j^{\rm eq}\times\mathbf{l}_{jb} = 0, \tag{3.1b}$$

so that $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ is constant.
BF angular velocity:

$$(\mathbf{S}^T\dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma}\,\omega_\gamma \Longrightarrow \mathbf{S}^T\dot{\mathbf{S}}\,\mathbf{v} = \boldsymbol{\omega}\times\mathbf{v}. \tag{3.2}$$

Velocities (using $|\mathbf{S}\mathbf{v}|^2 = |\mathbf{v}|^2$):

$$\dot{\mathbf{R}}'_j = \mathbf{S}\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big), \qquad \dot{\mathbf{r}}'_i = \mathbf{S}\big(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\big). \tag{3.3}$$

3.2 From the internal Hamiltonian to a working Routhian

I need an action principle for the body-fixed *nuclear* coordinates. The electrons, however, are best left in the canonical (momentum) form they already carry in (2.13): their kinetic energy and mass polarization form a rotational scalar that passes through the body-fixed transformation unchanged. I therefore Legendre-transform *only* the COM and nuclear momenta back to velocities — keeping $\mathbf{P}_{\text{cm}}$ arbitrary, never setting it to zero — and keep the electron momenta $\mathbf{p}'_i$ as canonical variables. The relevant Hamilton equations from (2.13) are

$$\dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}}/M_{\text{tot}}, \qquad \dot{\mathbf{R}}'_j = \mathbf{P}'_j/M_j, \qquad \dot{\mathbf{r}}'_i = \mathbf{p}'_i/m_e + (\textstyle\sum_l \mathbf{p}'_l)/M_N, \tag{3.4}$$

the last of which inverts to the electron momentum–velocity relation [recorded for the velocity-form cross-check below, around (3.18)]

$$\mathbf{p}'_i = m_e\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\text{tot}}}\sum_l \dot{\mathbf{r}}'_l, \qquad \sum_l \mathbf{p}'_l = \frac{m_e M_N}{M_{\text{tot}}}\sum_i \dot{\mathbf{r}}'_i. \tag{3.5}$$

Legendre-transforming the COM and nuclear momenta only, $\mathcal{R} = \mathbf{P}_{\text{cm}}\cdot\dot{\mathbf{R}}_{\text{cm}} + \sum_j \mathbf{P}'_j\cdot\dot{\mathbf{R}}'_j - H$, yields the **Routhian**

$$\boxed{\mathcal{R} = \tfrac{1}{2}M_{\text{tot}}|\dot{\mathbf{R}}_{\text{cm}}|^2 + \tfrac{1}{2}\sum_j M_j|\dot{\mathbf{R}}'_j|^2 - H_e(\mathbf{p}'_i) - V, \qquad H_e \equiv \sum_i \frac{\mathbf{p}_i'^{\,2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2,} \tag{3.6}$$

which is a Lagrangian in $(\mathbf{R}_{\text{cm}}, \mathbf{R}'_j)$ and *minus* a Hamiltonian in $(\mathbf{r}'_i, \mathbf{p}'_i)$: extremizing $\int \mathcal{R}\,dt$ gives Euler–Lagrange equations for the COM and nuclei and Hamilton's equations for the electrons. The electronic kinetic energy and mass polarization thus retain their (2.13) momentum form H_e — they are never converted to velocity form. (The overall minus sign on H_e is the Routhian's partial-Legendre signature; the mass polarization keeps its physical positive coefficient $+1/2M_N$ inside H_e, and the whole H_e re-enters the Hamiltonian with a plus sign when it is reassembled in (3.22).)

3.3 Body-fixed Routhian

Substitute the body-fixed velocities (3.3) into the nuclear term of (3.6); the COM term is untouched by the rotation. The electron Hamiltonian H_e is a *rotational scalar* — it is built from the dot products $\mathbf{p}'_i\cdot\mathbf{p}'_l$ — so under the passive rotation to the body frame it is unchanged in form. Defining the body-fixed electron momenta $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T\mathbf{p}'_i$ (a proper rotation, so $|\bar{\mathbf{p}}_i|^2 = |\mathbf{p}'_i|^2$ and $\sum_i \bar{\mathbf{p}}_i = \mathbf{S}^T\sum_i \mathbf{p}'_i$, whence $(\sum_i \bar{\mathbf{p}}_i)^2 = (\sum_i \mathbf{p}'_i)^2$),

$$H_e = \sum_i \frac{\mathbf{p}_i'^{\,2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2 = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2. \tag{3.6a}$$

Defining $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$, the body-fixed Routhian is therefore

$$\mathcal{R} = \tfrac{1}{2} M_{\text{tot}} |\dot{\mathbf{R}}_{\text{cm}}|^2 + \tfrac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N} \Big(\sum_i \bar{\mathbf{p}}_i \Big)^2 - V, \tag{3.7}$$

in which only the nuclear kinetic term carries the body-fixed angular velocity $\boldsymbol{\omega}$. The potential is now manifestly $\mathbf{S}$-independent:

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{j \neq k} \frac{Z_j Z_k e^2}{\bar{R}_{jk}} - \sum_{j,i} \frac{Z_j e^2}{|\bar{\mathbf{R}}_j - \bar{\mathbf{r}}_i|} + \frac{1}{2} \sum_{i \neq l} \frac{e^2}{\bar{r}_{il}} \right). \tag{3.8}$$

Rotational symmetry is rendered explicit; the BF electronic problem at fixed Q is self-contained.

3.4 Expanding (3.7)

3.4.1 Generic squared-term identity

For any vectors $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega} \in \mathbb{R}^3$, the BCH-style identity $\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega}\, a^2 - \mathbf{a}(\mathbf{a} \cdot \boldsymbol{\omega})$ gives

$$\begin{aligned} |\boldsymbol{\omega} \times \mathbf{a}|^2 = (\boldsymbol{\omega} \times \mathbf{a}) \cdot (\boldsymbol{\omega} \times \mathbf{a}) \;&=\; \boldsymbol{\omega} \cdot \big[\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a})\big] \\ = \boldsymbol{\omega} \cdot \big[\boldsymbol{\omega}\, a^2 - \mathbf{a}(\mathbf{a} \cdot \boldsymbol{\omega})\big] \;&=\; \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a}\mathbf{a}^T)\, \boldsymbol{\omega}, \end{aligned} \tag{3.9a}$$

while the scalar triple product gives

$$(\boldsymbol{\omega} \times \mathbf{a}) \cdot \dot{\mathbf{a}} \;=\; \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}}). \tag{3.9b}$$

Hence, for any constant mass-like coefficient m,

$$\begin{aligned} \tfrac{m}{2} \big|\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}\big|^2 &= \tfrac{m}{2} \big[|\boldsymbol{\omega} \times \mathbf{a}|^2 + 2(\boldsymbol{\omega} \times \mathbf{a}) \cdot \dot{\mathbf{a}} + |\dot{\mathbf{a}}|^2\big] \\ &= \underbrace{\tfrac{m}{2}\, \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a}\mathbf{a}^T) \boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m\, \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}})}_{\text{Coriolis-type}} + \underbrace{\tfrac{m}{2} |\dot{\mathbf{a}}|^2}_{\text{vibrational}}. \end{aligned} \tag{3.9c}$$

3.4.2 Application to the nuclear term

Only the nuclear kinetic term of the Routhian (3.7) carries $\boldsymbol{\omega}$. Apply (3.9c) with $m \to M_j$, $\mathbf{a} \to \bar{\mathbf{R}}_j$ and sum over j:

$$\tfrac{1}{2} \sum_j M_j \big|\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big|^2 \;=\; \tfrac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} \;+\; \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} \;+\; T_{\text{vib}}^N. \tag{3.9d}$$

The electronic kinetic energy is *not* expanded this way — it is carried as the rotational scalar H_e of (3.6a). Only for the velocity-form cross-check of § 3.6 do we record the decomposition that the electronic and mass-polarization terms *would* produce if the electrons were instead kept in velocity form,

$$\tfrac{m_e}{2} \sum_i \big|\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i\big|^2 = \tfrac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_e \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_e + T_e^{\text{BF}}, \qquad -\tfrac{m_e^2}{2M_{\text{tot}}} \big|\boldsymbol{\omega} \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}\big|^2 = -\tfrac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_X \boldsymbol{\omega} - \boldsymbol{\omega} \cdot \mathbf{L}_X - T_X^{\text{mp}}. \tag{3.9e}$$

The constituent inertia tensors are

$$\mathbf{I}_N = \sum_j M_j(\bar{R}_j^2 \mathbb{1} - \bar{\mathbf{R}}_j \bar{\mathbf{R}}_j^T), \qquad \mathbf{I}_e = m_e \sum_i (\bar{r}_i^2 \mathbb{1} - \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T), \tag{3.9}$$

$$\mathbf{I}_X = \tfrac{m_e^2}{M_{\text{tot}}}(\bar{r}^2 \mathbb{1} - \bar{\mathbf{r}}\bar{\mathbf{r}}^T), \tag{3.10}$$

the angular-momentum-like (velocity-form) quantities are

$$\mathbf{L}_N^{\text{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j, \quad \mathbf{L}_e = m_e \sum_i \bar{\mathbf{r}}_i \times \dot{\bar{\mathbf{r}}}_i, \quad \mathbf{L}_X = \tfrac{m_e^2}{M_{\text{tot}}} \bar{\mathbf{r}} \times \dot{\bar{\mathbf{r}}}, \tag{3.11}$$

and the "vibrational" kinetic pieces are

$$T_{\text{vib}}^N = \tfrac{1}{2}\sum_j M_j |\dot{\bar{\mathbf{R}}}_j|^2 = \tfrac{1}{2}\sum_b \dot{Q}_b^2, \quad T_e^{\text{BF}} = \tfrac{m_e}{2}\sum_i |\dot{\bar{\mathbf{r}}}_i|^2, \quad T_X^{\text{mp}} = \tfrac{m_e^2}{2M_{\text{tot}}}|\dot{\bar{\mathbf{r}}}|^2. \tag{3.12}$$

3.4.3 Assembled Routhian

Inserting the nuclear expansion (3.9d) into (3.7), together with the decoupled COM term, the electron scalar H_e, and the potential:

$$\boxed{\mathcal{R} = \tfrac{1}{2}M_{\text{tot}}|\dot{\mathbf{R}}_{\text{cm}}|^2 + \underbrace{\tfrac{1}{2}\boldsymbol{\omega}^T \mathbf{I}_N\, \boldsymbol{\omega} + \boldsymbol{\omega}\cdot\mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N}_{\text{nuclear, velocity form}} - H_e \;-\; V.} \tag{3.13}$$

The rotational kernel is the **nuclear** inertia $\mathbf{I}_N$ *alone*: because the electrons are carried as the scalar H_e rather than expanded in $\boldsymbol{\omega}$, no electronic or mass-polarization inertia ($\mathbf{I}_e$, $\mathbf{I}_X$) enters the kernel. The electrons couple to rotation only *kinematically*, through their angular momentum, which we identify next. The body-fixed electron momenta and the **electronic angular momentum** are

$$\bar{\mathbf{p}}_i = \mathbf{S}^T \mathbf{p}_i', \qquad \mathbf{L}_{\text{elec}} \equiv \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i. \tag{3.14}$$

Remark. The electronic and mass-polarization inertia/angular-momentum quantities $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ of (3.9)–(3.11), and the velocity-form decomposition (3.9e), are needed *only* in the cross-check of § 3.6; they play no role in the Routhian (3.13) itself.

3.5 Canonical momenta and the total angular momentum

Compute the conjugate momenta from the Routhian $\mathcal{R}$. The cyclic COM coordinate gives the conserved total momentum $\mathbf{P}_{\text{cm}} = \partial\mathcal{R}/\partial\dot{\mathbf{R}}_{\text{cm}} = M_{\text{tot}}\dot{\mathbf{R}}_{\text{cm}}$; being decoupled, it contributes nothing to the internal momenta below. The electron momenta $\bar{\mathbf{p}}_i$ are already canonical [eq. (3.14)]; it remains to find the rotational and vibrational momenta, most transparently by differentiating the unexpanded nuclear term of (3.7).

3.5.1 Nuclear angular momentum $\mathbf{J}_N = \partial\mathcal{R}/\partial\boldsymbol{\omega}$

In the Routhian (3.13) only the nuclear rotational kernel and the nuclear Coriolis piece carry $\boldsymbol{\omega}$ — the electron scalar H_e and the COM term do not. The momentum conjugate to the body-fixed angular velocity is therefore the *nuclear* angular momentum

$$\mathbf{J}_N \;\equiv\; \frac{\partial \mathcal{R}}{\partial \boldsymbol{\omega}} = \mathbf{I}_N\,\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}}. \tag{3.15}$$

3.5.2 Total angular momentum $\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\mathrm{elec}}$

The body-fixed frame $\mathbf{S}(\Omega)$ rotates the nuclei *and* the electrons together [eq. (3.1)], so the total angular momentum conjugate to the Euler angles — the physical, conserved rovibronic angular momentum — is the sum of the nuclear and electronic contributions. The electronic part is the canonical electron angular momentum $\mathbf{L}_{\mathrm{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ of (3.14), the body-fixed image of $\sum_i \mathbf{r}'_i \times \mathbf{p}'_i$ (the cross product co-rotates with $\mathbf{S}$). Hence

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\mathrm{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}} + \mathbf{L}_{\mathrm{elec}}. \tag{3.16}$$

3.5.3 Vibrational momentum $P_b = \partial \mathcal{R}/\partial \dot{Q}_b$

Only $\dot{\bar{\mathbf{R}}}_j$ carries $\dot{Q}_b$ dependence; with $\partial \bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ from (3.1a),

$$P_b \;\equiv\; \frac{\partial \mathcal{R}}{\partial \dot{Q}_b} = \sum_j M_j \big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big)\cdot \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j}\,\big(\boldsymbol{\omega}\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j\big)\cdot \mathbf{l}_{jb}. \tag{3.17}$$

Separating the rotational and vibrational pieces defines the **Coriolis vectors $\boldsymbol{\sigma}_b$**,

$$\boldsymbol{\sigma}_b \equiv \sum_j M_j\, \bar{\mathbf{R}}_j \times \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j}\,\bar{\mathbf{R}}_j \times \mathbf{l}_{jb} = \sum_a \boldsymbol{\zeta}_{ab}\, Q_a, \qquad \boldsymbol{\zeta}_{ab} \equiv \sum_j \mathbf{l}_{ja}\times\mathbf{l}_{jb}, \tag{3.17a}$$

the second form following from (3.1a) with the Eckart condition $\sum_j \sqrt{M_j}\bar{\mathbf{R}}_j^{\mathrm{eq}}\times\mathbf{l}_{jb} = 0$. The vibrational metric is the identity, $\sum_j M_j(\partial\bar{\mathbf{R}}_j/\partial Q_a)\cdot(\partial\bar{\mathbf{R}}_j/\partial Q_b) = \sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}$, so the canonical vibrational momentum carries *no* Wilson G-matrix,

$$P_b \;=\; \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b(Q) \;+\; \dot{Q}_b. \tag{3.17b}$$

3.6 Key identity

3.6.1 The clean split

The total-angular-momentum decomposition (3.16) *is* the key identity: subtracting the electronic part from both sides,

$$\boxed{\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = \mathbf{I}_N(Q)\,\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}}.} \tag{3.19}$$

The right side is a purely nuclear-velocity quantity, with the *nuclear* inertia $\mathbf{I}_N$ as rotational kernel. The entire electronic and mass-polarization content sits on the left, in the kinematic shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\mathrm{elec}}$; it never enters the kernel, because in the Routhian (3.13) the electrons were carried as the scalar H_e and contributed no inertia at all.

3.6.2 Cross-check: the $\mathbf{I}_e/\mathbf{I}_X$ cancellation of the velocity-form route

Had we instead converted the electrons to velocity form — expanding their kinetic energy via (3.9e) — the electronic inertia and Coriolis terms would have entered *both* $\mathbf{J}$ and $\mathbf{L}_{\text{elec}}$, and (3.19) would emerge only after an exact cancellation. The velocity-form electron momentum is $\bar{\mathbf{p}}_i = m_e(\boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i) - \frac{m_e^2}{M_{\text{tot}}}(\boldsymbol{\omega}\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}})$ [the body-fixed image of (3.5)]; substituting into $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i\times\bar{\mathbf{p}}_i$ and using the BAC–CAB identity $\mathbf{a}\times(\boldsymbol{\omega}\times\mathbf{a}) = (a^2\mathbb{1} - \mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}$ on the two $\boldsymbol{\omega}$-terms collapses the four pieces to

$$\mathbf{L}_{\text{elec}} = (\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X), \tag{3.18}$$

with $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ from (3.9)–(3.11). The velocity-form total angular momentum would read $\mathbf{J} = (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X)$ [from (3.9d)+(3.9e)], so

$$\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \underbrace{\big[(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X)\big] - \mathbf{L}_{\text{elec}}}_{=0 \text{ by } (3.18)} = \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}},$$

reproducing (3.19): the electronic inertia $\mathbf{I}_e$ and the mass-polarization inertia $\mathbf{I}_X$ cancel *exactly*. This "algebraic miracle" of the Eckart construction is automatic in the mixed treatment, where $\mathbf{I}_e$ and $\mathbf{I}_X$ never arise.

3.6.3 Inversion: the Watson reciprocal inertia

To invert (3.19) for $\boldsymbol{\omega}$ we must re-express the velocity-form $\mathbf{L}_N^{\text{vib}}$ in canonical variables. In normal coordinates $\mathbf{L}_N^{\text{vib}} = \sum_j M_j\bar{\mathbf{R}}_j\times\dot{\bar{\mathbf{R}}}_j = \sum_b \boldsymbol{\sigma}_b\,\dot{Q}_b$; eliminating $\dot{Q}_b = P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b$ from (3.17b),

$$\mathbf{L}_N^{\text{vib}} = \sum_b \boldsymbol{\sigma}_b\big(P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b\big) = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\omega}, \qquad \boldsymbol{\pi} \equiv \sum_b \boldsymbol{\sigma}_b P_b, \quad \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}} \equiv \sum_b \boldsymbol{\sigma}_b\boldsymbol{\sigma}_b^{\mathsf{T}}, \tag{3.20a}$$

with $\boldsymbol{\pi}$ the **field-free vibrational angular momentum**. Substituting into (3.19), $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})\boldsymbol{\omega} + \boldsymbol{\pi} \equiv \mathbf{I}'\boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boldsymbol{\omega} = \boldsymbol{\mu}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big), \qquad \boxed{\boldsymbol{\mu} \equiv (\mathbf{I}')^{-1} = \big(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\big)^{-1},} \tag{3.20}$$

the **Watson reciprocal-inertia tensor** [3]. The rotation–vibration coupling is now carried by the modified inertia $\mathbf{I}'$ and the uniform shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}$ — with no Wilson G-matrix.

3.7 Hamiltonian in canonical variables

We now assemble the body-fixed Hamiltonian in the canonical variables $(\Omega, \mathbf{J};\, Q, P_b;\, \bar{\mathbf{r}}_i, \bar{\mathbf{p}}_i)$. The total energy splits as $H = \mathbf{P}_{\text{cm}}^2/2M_{\text{tot}} + H_{\text{int}}$, the COM piece being a decoupled spectator. The internal Hamiltonian is the sum of the nuclear rotation–vibration kinetic energy, the electron scalar, and the potential,

$$H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V, \qquad T_N^{\text{rot+vib}} \equiv \tfrac{1}{2}\boldsymbol{\omega}^T\mathbf{I}_N\boldsymbol{\omega} + \boldsymbol{\omega}\cdot\mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N,$$

read off the Routhian (3.13). The electron scalar H_e [eq. (3.6a)] and the potential V are already canonical; only $T_N^{\text{rot+vib}}$ must be re-expressed through the momenta $(\mathbf{J}, P_b)$.

3.7.1 Nuclear rotation–vibration block

$T_N^{\text{rot+vib}}$ is homogeneous-quadratic in the nuclear velocities $(\boldsymbol{\omega}, \dot{Q}_b)$, so by Euler's theorem $2T_N^{\text{rot+vib}} = \mathbf{J}_N \cdot \boldsymbol{\omega} + \sum_b P_b \dot{Q}_b$, with conjugate momenta $\mathbf{J}_N = \mathbf{J} - \mathbf{L}_{\text{elec}}$ [eqs. (3.15),(3.19)] and P_b [eq. (3.17b)]. Substituting $\dot{Q}_b = P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b$ and $\sum_b P_b \boldsymbol{\sigma}_b = \boldsymbol{\pi}$,

$$2T_N^{\text{rot+vib}} = (\mathbf{J} - \mathbf{L}_{\text{elec}})\cdot\boldsymbol{\omega} + \sum_b P_b\big(P_b - \boldsymbol{\omega}\cdot\boldsymbol{\sigma}_b\big) = (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})\cdot\boldsymbol{\omega} + \sum_b P_b^2.$$

Inserting $\boldsymbol{\omega} = \boldsymbol{\mu}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})$ from (3.20),

$$\boxed{T_N^{\text{rot+vib}}\Big|_{\text{canonical}} = \tfrac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}\,(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \tfrac{1}{2}\sum_b P_b^2.} \tag{3.20d}$$

The vibrational kinetic energy is the *bare* $\frac{1}{2}\sum_b P_b^2$; the entire rotation–vibration coupling is carried by $\boldsymbol{\pi}$ and by the Watson inertia $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$ — no Wilson G-matrix survives.

3.7.2 Electronic block

The electronic kinetic energy and mass polarization require no further work: they are the rotational scalar H_e of (3.6a), carried unchanged from the internal Hamiltonian (2.13),

$$\boxed{H_e = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2.} \tag{3.21}$$

Because H_e is built from the rotation-invariant dot products $\bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_l$ and $\bar{\mathbf{p}}_i = \mathbf{S}^T \mathbf{p}'_i$ is a passive rotation, its body-fixed form is *identical* to the internal Hamiltonian's form (2.13) under $\mathbf{p}'_i \to \bar{\mathbf{p}}_i$ — no Legendre transform or angular-momentum expansion is needed. The mass polarization survives the body-fixed transformation intact in functional form.

Reduced-mass check ($N_{\text{elec}} = 1$): $H_e = \bar{\mathbf{p}}_1^2/(2m_e) + \bar{\mathbf{p}}_1^2/(2M_N) = \bar{\mathbf{p}}_1^2/(2\mu_e)$ with $\mu_e = m_e M_N/(m_e + M_N) = m_e M_N/M_{\text{tot}}$. ✓

3.7.3 Final assembly: (3.22)

Add (3.20d) and (3.21), include $V = V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$ from (3.8), and use $H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V$:

$$\boxed{\begin{aligned} H_{\text{int}} &= \tfrac{1}{2}\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big)^T \boldsymbol{\mu}(Q)\big(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}\big) \\ &\quad + \tfrac{1}{2}\sum_b P_b^2 \\ &\quad + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2 \\ &\quad + V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\}). \end{aligned}} \tag{3.22}$$

Restoring the decoupled COM piece, the complete canonical Hamiltonian is $H = \mathbf{P}_{\rm cm}^2/2M_{\rm tot} + H_{\rm int}$ with $H_{\rm int}$ given by (3.22) — exact for *any* $\mathbf{P}_{\rm cm}$. The COM term is a conserved free-particle spectator that decouples from every body-fixed variable, so the quantization of Section 4 proceeds with $H_{\rm int}$.

Here $\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b$ is the field-free vibrational angular momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ of (3.17a) and constants $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$; the vibrational metric being the identity, no Wilson G-matrix enters.

$\mathbf{L}_{\rm elec} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ is the BF canonical electronic angular momentum, and crucially:

- $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$ is the **Watson reciprocal-inertia tensor**, built from nuclear quantities alone: *no* electron or mass-polarization corrections enter the rotational kernel — they were absorbed into $\mathbf{L}_{\rm elec}$ via (3.19).
- The electron contribution to rotation appears entirely as the Coriolis-like coupling $-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\rm elec}$.
- The mass polarization is preserved in its $\left(\sum_i \bar{\mathbf{p}}_i\right)^2/(2M_N)$ form.

3.8 Complete coupling enumeration

Expanding the rotational kernel:

$$\tfrac{1}{2}\big(\mathbf{J} - \mathbf{L}_{\rm elec} - \boldsymbol{\pi}\big)^T \boldsymbol{\mu} \big(\mathbf{J} - \mathbf{L}_{\rm elec} - \boldsymbol{\pi}\big) =$$

$$\underbrace{\tfrac{1}{2}\mathbf{J}^T\boldsymbol{\mu}\mathbf{J}}_{\text{rotation}} \;\underbrace{-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}}_{\text{rot–vib Coriolis}} \;\underbrace{-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\rm elec}}_{\text{rot–elec Coriolis}} \;\underbrace{+\boldsymbol{\pi}^T\boldsymbol{\mu}\mathbf{L}_{\rm elec}}_{\text{vib–elec angular}} + \tfrac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi} + \tfrac{1}{2}\mathbf{L}_{\rm elec}^T\boldsymbol{\mu}\mathbf{L}_{\rm elec}.$$

All couplings in (3.22), referred to the standard analyses [3, 4]:

1. **Pure rotation, anisotropic top:** $\tfrac{1}{2}\mathbf{J}^T\boldsymbol{\mu}(Q)\mathbf{J}$ — Q-dependent moments of inertia (rot–vib coupling through inertia).
2. **Rot–vib Coriolis** [5]: $-\mathbf{J}^T\boldsymbol{\mu}\boldsymbol{\pi}$.
3. **Rot–electron Coriolis:** $-\mathbf{J}^T\boldsymbol{\mu}\mathbf{L}_{\rm elec}$ — the kinetic source of Λ-doubling [6, 7], Renner–Teller [8], and spin–rotation [9] analogues.
4. **Vib–electron angular coupling:** $\boldsymbol{\pi}^T\boldsymbol{\mu}\mathbf{L}_{\rm elec}$ — exact non-adiabatic angular coupling intrinsic to BF embedding [10, 11, 12].
5. **Quadratic vibrational angular momentum:** $\tfrac{1}{2}\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi}$.
6. **Quadratic electronic angular momentum:** $\tfrac{1}{2}\mathbf{L}_{\rm elec}^T\boldsymbol{\mu}\mathbf{L}_{\rm elec}$.
7. **Pure vibrational kinetic:** $\tfrac{1}{2}\sum_b P_b^2$.
8. **Electronic kinetic:** $\sum_i \bar{\mathbf{p}}_i^2/(2m_e)$.
9. **Mass-polarization, diagonal:** $\sum_i \bar{\mathbf{p}}_i^2/(2M_N)$ — modifies effective electron mass [10, 4].
10. **Mass-polarization, off-diagonal (two-electron):** $\sum_{i<l} \bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_l / M_N$.

11. **Coulomb:** $V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$. All Euler-angle independent.

Compared to the previous (electrons-in-SF) framework: kinetic rot–electron coupling has *replaced* potential rot–electron coupling. V_{Ne} is now manifestly $\mathbf{S}$-independent — the canonical chemistry advantage.

4 Quantization

The classical internal Hamiltonian (3.22) is parametrized by curvilinear coordinates (Euler angles ϕ, θ, χ and vibrational normal coordinates Q_b) and BF Cartesian electron coordinates $\bar{\mathbf{r}}_i$. Canonical quantization in curvilinear coordinates requires care: the naïve substitution $\pi_A \to -i\hbar\,\partial_A$ does not produce a Hermitian operator on the natural Hilbert space. The Podolsky (Laplace–Beltrami) prescription [13] resolves this, and we carry it out here *entirely in coordinate form*: we first construct the metric tensor of the ro-vibrational configuration space, then its determinant (which fixes the volume element) and its inverse, and then apply the Laplace–Beltrami operator block by block, writing out the explicit derivatives $\partial_\phi, \partial_\theta, \partial_\chi, \partial_{Q_b}$ at every stage. Hermiticity is automatic at each step because the Laplace–Beltrami operator is Hermitian by construction; in particular, *no commutator algebra of body-fixed angular momenta is ever used* (see the Remark in § 4.11). The residual ordering effects collapse into a single multiplicative function of the shape — the Watson pseudopotential. The term-by-term expansion of every block, with each derivative written out, is collected in Appendix B.

4.1 Why curvilinear coordinates require care

Label the curvilinear configuration-space coordinates

$$\{q^A\} \;=\; \big\{\,\phi, \theta, \chi,\; Q_1, \ldots, Q_f\,\big\}, \qquad f \equiv 3N_{\text{nucl}} - 6, \tag{4.1}$$

supplemented by the BF Cartesian electron coordinates $\bar{\mathbf{r}}_i$ (treated in § 4.4). In classical mechanics, any kinetic energy that is quadratic in the velocities defines a **metric tensor** $g_{AB}(q)$ on configuration space,

$$T = \tfrac{1}{2}\, g_{AB}(q)\, \dot{q}^A \dot{q}^B, \qquad p_A \equiv \frac{\partial T}{\partial \dot{q}^A} = g_{AB}\, \dot{q}^B, \tag{4.2}$$

(repeated indices summed), and eliminating the velocities in favor of the momenta gives the Hamiltonian form

$$T = \tfrac{1}{2}\, g^{AB}(q)\, p_A p_B, \qquad g^{AB} \equiv (g^{-1})_{AB}. \tag{4.3}$$

So three objects control everything: the metric g_{AB}, its inverse g^{AB}, and its determinant $g \equiv \det g_{AB}$. We will build all three explicitly for the coordinates (4.1) in § 4.2–4.6.

Why can we not simply replace $p_A \to -i\hbar\,\partial_A$ in (4.3)? Because on a curved configuration space the physically meaningful inner product carries the Riemannian volume weight, $\langle\varphi|\psi\rangle = \int \varphi^*\psi\,\sqrt{g}\,d^nq$, and $-i\hbar\partial_A$ is *not* Hermitian with respect to it. The one-dimensional case shows the problem completely: on $L^2(w(q)\,dq)$ with a positive weight w, integration by parts (boundary terms vanishing) gives

$$\int \varphi^* \big(-i\hbar\,\partial_q\psi\big)\, w\, dq = \int \big(-i\hbar\,\partial_q\varphi\big)^* \psi\, w\, dq \;-\; i\hbar \int \varphi^*\psi\, \frac{w'}{w}\, w\, dq, \tag{4.4}$$

so the adjoint is $(-i\hbar\partial_q)^\dagger = -i\hbar\big(\partial_q + w'/w\big) \neq -i\hbar\partial_q$ whenever the weight is not constant. A kinetic operator built by naïve substitution would fail to be Hermitian and could produce complex expectation values.

The resolution, due to Podolsky [13], is to replace the classical kinetic energy (4.3) by the **Laplace–Beltrami operator**:

$$\boxed{\hat{T} \;=\; -\frac{\hbar^2}{2}\,\frac{1}{\sqrt{g}}\,\partial_A\!\big(\sqrt{g}\,g^{AB}\,\partial_B\big), \qquad g=\det g_{AB}.} \tag{4.5}$$

This operator is Hermitian on $L^2(\sqrt{g}\,d^nq)$ *by construction*: for wavefunctions with vanishing boundary terms (here: periodicity in ϕ,χ, the vanishing of $\sin\theta$ at the poles $\theta=0,\pi$, and decay in Q_b), one integration by parts gives

$$\int \varphi^*\,\hat{T}\psi\;\sqrt{g}\,d^nq = \frac{\hbar^2}{2}\int (\partial_A\varphi)^*\,g^{AB}\,(\partial_B\psi)\;\sqrt{g}\,d^nq, \tag{4.6}$$

which is manifestly symmetric under $\varphi\leftrightarrow\psi$ (and manifestly positive for $\varphi=\psi$, since g^{AB} is positive definite wherever the classical kinetic energy is). When the metric is constant and Cartesian, $g_{AB}=\delta_{AB}$, (4.5) reduces to the familiar $-\frac{\hbar^2}{2}\nabla^2$ — the prescription contains ordinary quantum mechanics as a special case. Our entire task in this section is therefore: *compute g_{AB}, g, and g^{AB} for the coordinates (4.1), then expand (4.5).*

4.2 Body-frame kinematics: the Euler-angle derivatives, explicitly

Fix the body-fixed orientation by the zyz convention

$$\mathbf{S}(\phi,\theta,\chi)=R_z(\phi)\,R_y(\theta)\,R_z(\chi), \qquad R_z(\psi)=\begin{pmatrix}\cos\psi & -\sin\psi & 0\\ \sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{pmatrix},\; R_y(\vartheta)=\begin{pmatrix}\cos\vartheta & 0 & \sin\vartheta\\ 0 & 1 & 0\\ -\sin\vartheta & 0 & \cos\vartheta\end{pmatrix}, \tag{4.7}$$

so that $\mathbf{R}'_j=\mathbf{S}\bar{\mathbf{R}}_j$ as in (3.1). The body-fixed angular velocity is defined by $(\mathbf{S}^T\dot{\mathbf{S}})_{\alpha\beta}=-\epsilon_{\alpha\beta\gamma}\omega_\gamma$ [eq. (3.2)]; our first job is to express $\boldsymbol{\omega}$ through $(\dot\phi,\dot\theta,\dot\chi)$. Differentiating the one-parameter rotations at zero angle defines the generators

$$G_z\equiv\partial_\psi R_z\big|_0=\begin{pmatrix}0 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad G_y\equiv\partial_\vartheta R_y\big|_0=\begin{pmatrix}0 & 0 & 1\\ 0 & 0 & 0\\ -1 & 0 & 0\end{pmatrix}, \tag{4.8}$$

with $\partial_\psi R_z(\psi)=G_zR_z(\psi)$ and $\partial_\vartheta R_y(\vartheta)=G_yR_y(\vartheta)$ (check by direct differentiation of (4.7)). The chain rule applied to (4.7) then gives $\dot{\mathbf{S}}=\dot\phi\,G_z\mathbf{S}+\dot\theta\,R_z(\phi)G_yR_y(\theta)R_z(\chi)+\dot\chi\,\mathbf{S}G_z$, and multiplying by $\mathbf{S}^T$ from the left,

$$\mathbf{S}^T\dot{\mathbf{S}}=\dot\phi\,\mathbf{S}^TG_z\mathbf{S}+\dot\theta\,R_z(\chi)^TG_yR_z(\chi)+\dot\chi\,G_z=\big[\dot\phi\,\mathbf{S}^T\hat{\mathbf{z}}+\dot\theta\,R_z(\chi)^T\hat{\mathbf{y}}+\dot\chi\,\hat{\mathbf{z}}\big]_\times, \tag{4.9}$$

where $[\mathbf{a}]_\times$ denotes the antisymmetric matrix with $([\mathbf{a}]_\times)_{\alpha\beta}=-\epsilon_{\alpha\beta\gamma}a_\gamma$, and we used the conjugation rule $R^T[\mathbf{a}]_\times R=[R^T\mathbf{a}]_\times$ for any rotation R, together with $R_y(\theta)^T\hat{\mathbf{y}}=\hat{\mathbf{y}}$. Evaluating the two nontrivial vectors from (4.7), $\mathbf{S}^T\hat{\mathbf{z}}=(-\sin\theta\cos\chi,\ \sin\theta\sin\chi,\ \cos\theta)^T$ and $R_z(\chi)^T\hat{\mathbf{y}}=(\sin\chi,\ \cos\chi,\ 0)^T$, and comparing with the definition of $\boldsymbol{\omega}$, we obtain $\boldsymbol{\omega}=\mathsf{E}(\Omega)\,\dot{\boldsymbol{\Omega}}$

with $\dot{\boldsymbol{\Omega}} = (\dot\phi, \dot\theta, \dot\chi)^T$ and the **kinematic matrix**

$$\boxed{\mathsf{E}(\Omega) = \begin{pmatrix} -\sin\theta\cos\chi & \sin\chi & 0 \\ \sin\theta\sin\chi & \cos\chi & 0 \\ \cos\theta & 0 & 1 \end{pmatrix}, \qquad \begin{aligned} \omega_1 &= -\dot\phi\,\sin\theta\cos\chi + \dot\theta\,\sin\chi, \\ \omega_2 &= \;\;\dot\phi\,\sin\theta\sin\chi + \dot\theta\,\cos\chi, \\ \omega_3 &= \;\;\dot\phi\,\cos\theta + \dot\chi. \end{aligned}} \tag{4.10}$$

Expanding the determinant along the third column,

$$\det\mathsf{E} = 1\cdot\det\begin{pmatrix} -\sin\theta\cos\chi & \sin\chi \\ \sin\theta\sin\chi & \cos\chi \end{pmatrix} = -\sin\theta\,(\cos^2\chi + \sin^2\chi) = -\sin\theta, \qquad |\det\mathsf{E}| = \sin\theta. \tag{4.11}$$

Solving the linear system (4.10) for the Euler rates — multiply the ω_1, ω_2 lines by $(-\cos\chi, \sin\chi)$ and by $(\sin\chi, \cos\chi)$ respectively and add, then substitute into the ω_3 line — gives $\dot{\boldsymbol{\Omega}} = \mathsf{E}^{-1}\boldsymbol{\omega}$ with

$$\boxed{\mathsf{E}^{-1}(\Omega) = \begin{pmatrix} -\dfrac{\cos\chi}{\sin\theta} & \dfrac{\sin\chi}{\sin\theta} & 0 \\ \sin\chi & \cos\chi & 0 \\ \dfrac{\cos\theta\cos\chi}{\sin\theta} & -\dfrac{\cos\theta\sin\chi}{\sin\theta} & 1 \end{pmatrix}, \qquad \begin{aligned} \dot\phi &= \tfrac{1}{\sin\theta}\big(-\cos\chi\,\omega_1 + \sin\chi\,\omega_2\big), \\ \dot\theta &= \sin\chi\,\omega_1 + \cos\chi\,\omega_2, \\ \dot\chi &= \omega_3 + \tfrac{\cos\theta}{\sin\theta}\big(\cos\chi\,\omega_1 - \sin\chi\,\omega_2\big). \end{aligned}} \tag{4.12}$$

The $1/\sin\theta$ entries reflect the coordinate singularity of the Euler parametrization at the poles $\theta = 0, \pi$ (where ϕ and χ describe the same rotation); it is compensated by the vanishing of $|\det\mathsf{E}| = \sin\theta$ in the volume element, and no physical quantity is singular there. Equations (4.10)–(4.12) are the only facts about the Euler angles used in the rest of this section.

4.3 The ro-vibrational metric tensor

The classical rotation–vibration kinetic energy in the body frame is the nuclear term of the Routhian (3.13),

$$2T_{\rm rv} = \boldsymbol{\omega}^T\mathbf{I}_N(Q)\,\boldsymbol{\omega} + 2\,\boldsymbol{\omega}^T\sum_b \boldsymbol{\sigma}_b(Q)\,\dot Q_b + \sum_b \dot Q_b^2, \tag{4.13}$$

whose three pieces were derived in § 3.4: the nuclear inertia tensor $\mathbf{I}_N = \sum_j M_j(\bar R_j^2\mathbb{1} - \bar{\mathbf{R}}_j\bar{\mathbf{R}}_j^T)$ [(3.9)], the Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab}Q_a$ with *constant* coefficients $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja}\times\mathbf{l}_{jb}$ [(3.17a)], and the identity vibrational metric $\sum_j M_j(\partial\bar{\mathbf{R}}_j/\partial Q_a)\cdot(\partial\bar{\mathbf{R}}_j/\partial Q_b) = \delta_{ab}$, a consequence of the mass-weighted rectilinear normal coordinates [(3.17)]. Collect the Coriolis vectors into the $3\times f$ matrix $\sigma_{\alpha b} \equiv (\boldsymbol{\sigma}_b)_\alpha$ and write (4.13) as one quadratic form in the *physical* velocities,

$$2T_{\rm rv} = \begin{pmatrix} \boldsymbol{\omega} \\ \dot{\mathbf{Q}} \end{pmatrix}^T \begin{pmatrix} \mathbf{I}_N & \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T & \mathbb{1}_f \end{pmatrix} \begin{pmatrix} \boldsymbol{\omega} \\ \dot{\mathbf{Q}} \end{pmatrix}. \tag{4.14}$$

The metric of (4.2), however, must be expressed in the *coordinate* velocities $\dot q = (\dot\phi, \dot\theta, \dot\chi, \dot Q_b)$. The two are related linearly by (4.10): $\binom{\boldsymbol{\omega}}{\dot{\mathbf{Q}}} = \mathsf{A}\,\dot q$ with $\mathsf{A} = \mathrm{diag}(\mathsf{E}, \mathbb{1}_f)$. Substituting into

(4.14), the metric g_{AB} defined by $2T_{\rm rv} = g_{AB}\dot q^A \dot q^B$ is $g = \mathsf{A}^T(\cdots)\mathsf{A}$:

$$\boxed{g_{AB} = \begin{pmatrix} \mathsf{E}^T\mathbf{I}_N\,\mathsf{E} & \mathsf{E}^T\boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T\mathsf{E} & \mathbb{1}_f \end{pmatrix}, \qquad \begin{aligned} &\text{Euler–Euler block: } \mathsf{E}^T\mathbf{I}_N\mathsf{E} \quad (3\times 3), \\ &\text{Euler–mode block: } \mathsf{E}^T\boldsymbol{\sigma} \quad (3\times f), \\ &\text{mode–mode block: } \mathbb{1}_f. \end{aligned}} \tag{4.15}$$

Every entry is now an explicit function of (θ, χ, Q): the inertia tensor pulled back through the kinematic matrix, the Coriolis coupling pulled back through the same matrix, and a flat vibrational block. This single object encodes all four kinetic structures of the classical Hamiltonian (3.22) — the rotational kernel, the Coriolis coupling, the vibrational kinetic energy, and the internal $\boldsymbol{\pi}^T\boldsymbol{\mu}\boldsymbol{\pi}$ term — as we verify in § 4.7 after inverting it.

4.4 The electron sector: flat, constant, and decoupled from the measure

Before quantizing (4.15) we must account for the electrons, since the Podolsky prescription requires the metric of the *full* configuration space. Include the electron velocities: expanding the body-fixed electronic and mass-polarization kinetic terms via (3.9e), the complete velocity-form kinetic energy is

$$2T = \mathbf{v}^{\mathsf{T}}\mathsf{M}\,\mathbf{v}, \qquad \mathbf{v} = (\boldsymbol{\omega}, \dot{\mathbf{Q}}, \dot{\bar{\mathbf{r}}}_i), \qquad \mathsf{M} = \begin{pmatrix} \mathbf{I} & \boldsymbol{\sigma} & \mathsf{C} \\ \boldsymbol{\sigma}^{\mathsf{T}} & \mathbb{1}_f & 0 \\ \mathsf{C}^{\mathsf{T}} & 0 & \mathsf{D} \end{pmatrix}, \tag{4.16}$$

with $\mathbf{I} = \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ the full instantaneous inertia, C the rotation–electron coupling block [from $\boldsymbol{\omega}\cdot(\mathbf{L}_e - \mathbf{L}_X)$ in (3.9e)], and $\mathsf{D}_{(i\alpha)(j\beta)} = (m_e\delta_{ij} - m_e^2/M_{\rm tot})\,\delta_{\alpha\beta}$ the *constant* electron mass matrix. Eliminating the electron block by its Schur complement — i.e. completing the square in $\dot{\bar{\mathbf{r}}}_i$ — removes all $\boldsymbol{\omega}$-dependence from the electron sector: the correction is precisely the electronic-plus-mass-polarization inertia, $\mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_e - \mathbf{I}_X$, so $\mathbf{I} - \mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_N$ — the same exact cancellation proven classically in the cross-check of § 3.6 [eq. (3.18)]. Hence

$$\det\mathsf{M} = \det\mathsf{D}\ \det\begin{pmatrix} \mathbf{I}_N & \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^{\mathsf{T}} & \mathbb{1}_f \end{pmatrix}, \qquad \det\mathsf{D} = \text{const}, \tag{4.17}$$

and the surviving determinant is exactly that of the ro-vibrational block (4.14). Two consequences. *(i)* The electron coordinates contribute only a constant factor to the volume element — their measure is the flat $\prod_i d^3\bar{\mathbf{r}}_i$, and $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on it with no weight corrections. *(ii)* In the canonical (momentum) form (3.22), the electrons couple to rotation only through the shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\rm elec}$, and the operator $\hat{L}_{{\rm elec},\alpha} = -i\hbar\sum_i \epsilon_{\alpha\beta\gamma}\bar{r}_{i\beta}\partial_{\bar{r}_{i\gamma}}$ is Hermitian on the flat electron measure *without any symmetrization*, because its vector field is divergence-free:

$$\sum_{i\gamma}\partial_{\bar{r}_{i\gamma}}\big(\epsilon_{\alpha\beta\gamma}\bar{r}_{i\beta}\big) = \sum_i \epsilon_{\alpha\beta\gamma}\,\delta_{\beta\gamma} = 0. \tag{4.18}$$

Consequently the entire quantization problem lives in the ro-vibrational sector: we quantize the metric (4.15) by the Podolsky prescription, and the electrons enter the result only through the flat kinetic operator of § 4.10 and the uniform shift $\hat{J}_\alpha \to \hat{J}_\alpha - \hat{L}_{\mathrm{elec},\alpha}$, which preserves Hermiticity term by term because $\hat{L}_{\mathrm{elec},\alpha}$ is Hermitian and acts on a different set of coordinates than every ro-vibrational object (the term-by-term verification is Appendix B).

4.5 Determinant and the volume element

We need $g = \det g_{AB}$ for the metric (4.15). Use the block-determinant identity: for any square blocks with invertible lower-right corner,

$$\det\begin{pmatrix} \mathsf{a} & \mathsf{b} \\ \mathsf{b}^T & \mathbb{1} \end{pmatrix} = \det\!\big(\mathsf{a} - \mathsf{b}\,\mathsf{b}^T\big), \tag{4.19}$$

which follows by block row reduction — subtracting $\mathsf{b} \times$ (bottom row) from the top row produces the block-triangular matrix $\left(\begin{smallmatrix} \mathsf{a}-\mathsf{b}\mathsf{b}^T & 0 \\ \mathsf{b}^T & \mathbb{1} \end{smallmatrix}\right)$, whose determinant is the product of the diagonal blocks. Applying (4.19) to (4.15) with $\mathsf{a} = \mathsf{E}^T \mathbf{I}_N \mathsf{E}$, $\mathsf{b} = \mathsf{E}^T \boldsymbol{\sigma}$:

$$\det g_{AB} = \det\!\big[\mathsf{E}^T(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)\mathsf{E}\big] = (\det \mathsf{E})^2\, \det \mathbf{I}' \implies \boxed{\det g_{AB} = \sin^2\theta\; \gamma(Q), \quad \gamma \equiv \det \mathbf{I}', \quad \mathbf{I}' \equiv \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T,} \tag{4.20}$$

using $\det(\mathsf{E}^T X \mathsf{E}) = (\det \mathsf{E})^2 \det X$ and (4.11). Remarkably, the determinant is governed by the *Watson-modified* inertia $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T$ of (3.20) — not the bare $\mathbf{I}_N$: the Coriolis coupling $\boldsymbol{\sigma}$ reduces the rotational determinant, and it is the same $\mathbf{I}'$ whose inverse $\boldsymbol{\mu} = (\mathbf{I}')^{-1}$ is the rotational kernel of (3.22). The Riemannian volume element is therefore

$$\boxed{\sqrt{g}\, d^{3+f}q = \underbrace{\sin\theta\, d\phi\, d\theta\, d\chi}_{\text{Haar measure on } SO(3)} \; \underbrace{\sqrt{\gamma(Q)}\, \textstyle\prod_b dQ_b}_{\text{rot.–vib. Jacobian}}, \qquad \sqrt{g} = \sin\theta\, \sqrt{\gamma(Q)},} \tag{4.21}$$

times the flat, constant electron factor $\prod_i d^3\bar{\mathbf{r}}_i$ from § 4.4. The two curved factors have transparent origins: $\sin\theta$ is the Euler-angle Jacobian $|\det \mathsf{E}|$ of (4.11), and $\sqrt{\gamma}$ is the shape-dependent part carrying the Coriolis-reduced inertia.

4.6 The inverse metric

The Podolsky operator (4.5) needs g^{AB}. We claim

$$\boxed{g^{AB} = \begin{pmatrix} \mathsf{E}^{-1}\boldsymbol{\mu}\,\mathsf{E}^{-T} & -\mathsf{E}^{-1}\boldsymbol{\mu}\,\boldsymbol{\sigma} \\ -\boldsymbol{\sigma}^T\boldsymbol{\mu}\,\mathsf{E}^{-T} & \mathbb{1}_f + \boldsymbol{\sigma}^T\boldsymbol{\mu}\,\boldsymbol{\sigma} \end{pmatrix}, \qquad \boldsymbol{\mu} \equiv (\mathbf{I}')^{-1} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1},} \tag{4.22}$$

with $\mathsf{E}^{-T} \equiv (\mathsf{E}^{-1})^T$ from (4.12). Rather than derive it, *verify* it — multiply (4.15) by (4.22) block by block, using only $\boldsymbol{\mu}\mathbf{I}' = \mathbb{1}$:

$$
\begin{aligned}
\text{(top-left)}\quad & \mathsf{E}^T\mathbf{I}_N\mathsf{E}\cdot\mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T} - \mathsf{E}^T\boldsymbol{\sigma}\cdot\boldsymbol{\sigma}^T\boldsymbol{\mu}\mathsf{E}^{-T} = \mathsf{E}^T(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)\boldsymbol{\mu}\,\mathsf{E}^{-T} = \mathsf{E}^T\mathsf{E}^{-T} = \mathbb{1}_3,\\
\text{(top-right)}\quad & -\mathsf{E}^T\mathbf{I}_N\mathsf{E}\cdot\mathsf{E}^{-1}\boldsymbol{\mu}\boldsymbol{\sigma} + \mathsf{E}^T\boldsymbol{\sigma}\,(\mathbb{1}_f + \boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma}) = \mathsf{E}^T\big[-(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)\boldsymbol{\mu}\boldsymbol{\sigma} + \boldsymbol{\sigma}\big] = \mathsf{E}^T(-\boldsymbol{\sigma}+\boldsymbol{\sigma}) = 0,\\
\text{(bottom-left)}\quad & \boldsymbol{\sigma}^T\mathsf{E}\cdot\mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T} - \mathbb{1}_f\cdot\boldsymbol{\sigma}^T\boldsymbol{\mu}\mathsf{E}^{-T} = 0,\\
\text{(bottom-right)}\quad & -\boldsymbol{\sigma}^T\mathsf{E}\cdot\mathsf{E}^{-1}\boldsymbol{\mu}\boldsymbol{\sigma} + \mathbb{1}_f + \boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma} = \mathbb{1}_f.
\end{aligned}
\tag{4.23}
$$

So $g_{AB}\,g^{BC} = \delta_A^{\ C}$, confirming (4.22). Note how the Watson reciprocal inertia $\boldsymbol{\mu}$ — introduced classically in (3.20) by inverting the momentum–velocity relation — reappears here as the geometric inverse of the metric: the two inversions are the same computation.

4.7 The rotational derivative operators (defined, not postulated)

Expanding the Podolsky operator will repeatedly produce the combination $(\mathsf{E}^{-1})_{s\alpha}\partial_{\Omega_s}$ (index s running over ϕ, θ, χ). It is convenient to give it a name:

$$
\boxed{\hat{J}_\alpha \equiv -i\hbar\sum_s(\mathsf{E}^{-1})_{s\alpha}\,\partial_{\Omega_s}: \qquad
\begin{aligned}
\hat{J}_1 &= -i\hbar\Big[-\tfrac{\cos\chi}{\sin\theta}\,\partial_\phi + \sin\chi\,\partial_\theta + \tfrac{\cos\theta\cos\chi}{\sin\theta}\,\partial_\chi\Big],\\
\hat{J}_2 &= -i\hbar\Big[\ \tfrac{\sin\chi}{\sin\theta}\,\partial_\phi + \cos\chi\,\partial_\theta - \tfrac{\cos\theta\sin\chi}{\sin\theta}\,\partial_\chi\Big],\\
\hat{J}_3 &= -i\hbar\,\partial_\chi,
\end{aligned}}
\tag{4.24}
$$

reading the rows of E^{-1} off (4.12). We stress the logic: (4.24) is a *definition* — shorthand for three explicit combinations of Euler-angle derivatives — and we will never compute a commutator $[\hat{J}_\alpha, \hat{J}_\beta]$ or use any angular-momentum algebra. The definition is natural because classically $J_\alpha = \partial T_{\mathrm{rv}}/\partial\omega_\alpha$ is conjugate to $\boldsymbol{\omega}$: from $\boldsymbol{\omega} = \mathsf{E}\dot{\boldsymbol{\Omega}}$ and the chain rule, $p_{\Omega_s} = \partial T/\partial\dot{\Omega}_s = (\partial\omega_\alpha/\partial\dot{\Omega}_s)\,J_\alpha = (\mathsf{E})_{\alpha s}J_\alpha$, i.e. $p_\Omega = \mathsf{E}^T\mathbf{J}$ and $J_\alpha = (\mathsf{E}^{-1})_{s\alpha}\,p_{\Omega_s}$ — (4.24) is its coordinate representation. As a consistency check, substituting $p_\Omega = \mathsf{E}^T\mathbf{J}$ and P_b into $\frac{1}{2}p_A g^{AB} p_B$ with (4.22) collapses all E factors,

$$
\tfrac{1}{2}\,p_A g^{AB} p_B = \tfrac{1}{2}\,\mathbf{J}^T\boldsymbol{\mu}\,\mathbf{J} - \mathbf{J}^T\boldsymbol{\mu}\,\boldsymbol{\pi} + \tfrac{1}{2}\,\boldsymbol{\pi}^T\boldsymbol{\mu}\,\boldsymbol{\pi} + \tfrac{1}{2}\sum_b P_b^2, \qquad \boldsymbol{\pi} \equiv \boldsymbol{\sigma}P = \sum_b \boldsymbol{\sigma}_b P_b, \tag{4.25}
$$

which is exactly the nuclear kernel of (3.22) [the electronic shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\mathrm{elec}}$ enters per § 4.4]. So the classical correspondence is verified before any quantization.

Two coordinate identities make the operators (4.24) as good as Cartesian momenta. First, a **divergence identity**: the weighted divergence of each column of E^{-1} vanishes.

Compute all three explicitly from (4.12):

$$\begin{aligned}
\alpha = 1: \quad & \sum_s \partial_{\Omega_s}\big(\sin\theta\,(\mathsf{E}^{-1})_{s1}\big) = \partial_\phi(-\cos\chi) + \partial_\theta(\sin\theta\sin\chi) + \partial_\chi(\cos\theta\cos\chi) \\
& \qquad = 0 + \cos\theta\sin\chi - \cos\theta\sin\chi = 0, \\
\alpha = 2: \quad & \sum_s \partial_{\Omega_s}\big(\sin\theta\,(\mathsf{E}^{-1})_{s2}\big) = \partial_\phi(\sin\chi) + \partial_\theta(\sin\theta\cos\chi) + \partial_\chi(-\cos\theta\sin\chi) \\
& \qquad = 0 + \cos\theta\cos\chi - \cos\theta\cos\chi = 0, \\
\alpha = 3: \quad & \sum_s \partial_{\Omega_s}\big(\sin\theta\,(\mathsf{E}^{-1})_{s3}\big) = \partial_\chi(\sin\theta) = 0.
\end{aligned} \tag{4.26}$$

Second, this identity makes each $\hat{J}_\alpha$ **Hermitian on the Haar measure** $\sin\theta\, d\phi\, d\theta\, d\chi$: integrating by parts (periodicity in ϕ, χ; $\sin\theta \to 0$ at the poles),

$$\int \varphi^*\, \hat{J}_\alpha \psi\, \sin\theta\, d\Omega = \int (\hat{J}_\alpha\varphi)^* \psi\, \sin\theta\, d\Omega + i\hbar \int \varphi^* \psi\, \Big[\underbrace{\textstyle\sum_s \partial_{\Omega_s}\big(\sin\theta(\mathsf{E}^{-1})_{s\alpha}\big)}_{=0 \text{ by } (4.26)}\Big] d\Omega, \tag{4.27}$$

so $\hat{J}_\alpha^\dagger = \hat{J}_\alpha$ with *no* symmetrization needed. In differential-geometric language, (4.26) says the vector fields generating body-frame rotations are divergence-free with respect to the Haar measure; it is the coordinate-level fact that replaces every appeal to angular-momentum algebra in what follows. [The same argument with the flat measure and (4.18) makes $\hat{L}_{\text{elec},\alpha}$ Hermitian, and the vibrational analogue appears in § 4.9.]

4.8 The Podolsky operator, block by block

Insert the measure (4.21) and the inverse metric (4.22) into the Podolsky operator (4.5) and split the double sum over A, B into the four blocks:

$$\hat{T}_{\text{rv}} = \underbrace{-\frac{\hbar^2}{2\sqrt{g}}\, \partial_{\Omega_s}\big(\sqrt{g}\, g^{\Omega\Omega}_{st} \partial_{\Omega_t}\big)}_{\hat{T}_{\text{rot}}} \underbrace{-\frac{\hbar^2}{2\sqrt{g}}\Big[\partial_{\Omega_s}\big(\sqrt{g}\, g^{\Omega Q}_{sb} \partial_{Q_b}\big) + \partial_{Q_b}\big(\sqrt{g}\, g^{Q\Omega}_{bs} \partial_{\Omega_s}\big)\Big]}_{\hat{T}_{\text{Cor}}} \underbrace{-\frac{\hbar^2}{2\sqrt{g}}\, \partial_{Q_b}\big(\sqrt{g}\, g^{QQ}_{bc} \partial_{Q_c}\big)}_{\hat{T}_{\text{vib}}}. \tag{4.28}$$

Rotational block. In $\hat{T}_{\text{rot}}$, the factors $\sqrt{\gamma}$, $\mu_{\alpha\beta}(Q)$ are independent of the Euler angles, so they pass through ∂_{Ω_s} and the $\sqrt{\gamma}$ cancels against the prefactor. Writing $g^{\Omega\Omega}_{st} = (\mathsf{E}^{-1})_{s\alpha}\mu_{\alpha\beta}(\mathsf{E}^{-1})_{t\beta}$ and expanding the outer derivative by the product rule,

$$\begin{aligned}
\hat{T}_{\text{rot}} &= -\frac{\hbar^2}{2}\,\mu_{\alpha\beta}\,\Big[\underbrace{(\mathsf{E}^{-1})_{s\alpha}\,\partial_{\Omega_s}\big((\mathsf{E}^{-1})_{t\beta}\,\partial_{\Omega_t}\big)}_{\text{both derivatives survive}} + \underbrace{\tfrac{1}{\sin\theta}\Big(\textstyle\sum_s \partial_{\Omega_s}\big(\sin\theta(\mathsf{E}^{-1})_{s\alpha}\big)\Big)}_{=0 \text{ by } (4.26)}(\mathsf{E}^{-1})_{t\beta}\,\partial_{\Omega_t}\Big] \\
&= \tfrac{1}{2}\sum_{\alpha\beta} \mu_{\alpha\beta}(Q)\, \hat{J}_\alpha \hat{J}_\beta,
\end{aligned} \tag{4.29}$$

by the definition (4.24). The divergence identity (4.26) has done all the work: the rotational block is *exactly* the symmetric kernel $\frac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta$, Hermitian as written [each $\hat{J}_\alpha$ is Hermitian by (4.27), $\mu_{\alpha\beta}$ is a real symmetric function of Q commuting with both, and taking the adjoint merely reverses the order, $(\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta)^\dagger = \mu_{\alpha\beta}\hat{J}_\beta\hat{J}_\alpha$, which upon relabeling the summed indices $\alpha \leftrightarrow \beta$ and using $\mu_{\beta\alpha} = \mu_{\alpha\beta}$ is the original operator], and it contributes *no* extra potential term. In the traditional treatment this last fact is derived from the anomalous commutator algebra; here it is simply the statement that a rigid rotor has no pseudopotential, verified by the three explicit derivative computations (4.26).

Coriolis block. In $\hat{T}_{\rm Cor}$, the first group carries ∂_{Ω_s}, which passes through the Q-functions $\mu, \boldsymbol{\sigma}, \sqrt{\gamma}$ and, by the same divergence identity (4.26), collapses onto $\hat{J}_\alpha$ acting from the left. The second group carries ∂_{Q_b}, which passes through $\mathsf{E}^{-1}(\Omega)$ and $\sin\theta$ but *acts* on $\sqrt{\gamma}$, μ, and $\boldsymbol{\sigma}$. The result (each step is one product rule; the full expansion is Appendix B) is

$$\hat{T}_{\rm Cor} = -\tfrac{1}{2}\Big[\hat{J}_\alpha\, \mu_{\alpha\beta}\, \hat{\pi}_\beta + \tfrac{1}{\sqrt{\gamma}}\, \hat{P}_b\, \sqrt{\gamma}\, \sigma_{\alpha b}\, \mu_{\alpha\beta}\, \hat{J}_\beta\Big], \qquad \hat{P}_b \equiv -i\hbar\, \partial_{Q_b}, \quad \hat{\pi}_\alpha \equiv \sum_b \sigma_{\alpha b}(Q)\, \hat{P}_b. \tag{4.30}$$

The two groups are adjoints of each other on the measure (4.21) [the $\gamma^{\pm 1/2}$ factors in the second group are precisely what integration by parts over $\sqrt{\gamma}\, dQ$ produces], so their sum is Hermitian — again automatically, with no ordering imposed by hand.

Vibrational block. With $g^{QQ}_{bc} = \delta_{bc} + (\boldsymbol{\sigma}^T \boldsymbol{\mu}\boldsymbol{\sigma})_{bc}$ and the $\sin\theta$ cancelling,

$$\hat{T}_{\rm vib} = -\frac{\hbar^2}{2}\, \frac{1}{\sqrt{\gamma}} \sum_{b,c} \partial_{Q_b}\Big(\sqrt{\gamma}\, \big[\delta_{bc} + (\boldsymbol{\sigma}^T \boldsymbol{\mu}\boldsymbol{\sigma})_{bc}\big]\, \partial_{Q_c}\Big). \tag{4.31}$$

Expanding the δ_{bc} part by one product rule exhibits the problem announced in (4.4): $-\frac{\hbar^2}{2}\gamma^{-1/2}\partial_{Q_b}(\gamma^{1/2}\partial_{Q_b}\psi) = \frac{1}{2}\hat{P}_b^2\psi - \frac{\hbar^2}{2}\big(\frac{1}{2}\partial_{Q_b}\ln\gamma\big)\partial_{Q_b}\psi$ — the bare $\hat{P}_b$ is not Hermitian on the curved measure $\sqrt{\gamma}\, dQ$, and the first-order remainder is the compensation. Removing these remainders once and for all is the job of the rescaling below.

4.9 Rescaling to the flat measure and the Watson pseudopotential

To work with the simpler measure $d\mu_W \equiv \sin\theta\, d\phi\, d\theta\, d\chi\, \prod_b dQ_b\, \prod_i d^3\bar{\mathbf{r}}_i$, absorb the shape factor of the volume element into the wavefunction:

$$\psi_W \equiv \gamma^{1/4}\, \psi \qquad \Longrightarrow \qquad |\psi|^2\, \sqrt{g}\, d^{3+f}q = |\psi_W|^2\, d\mu_W, \tag{4.32}$$

so $\psi_W \in L^2(d\mu_W)$, and conjugate every operator: $\hat{O}_W = \gamma^{1/4}\, \hat{O}\, \gamma^{-1/4}$. Only the Q-sector is affected (γ depends on Q alone); the $\sin\theta$ stays in the measure, on which the $\hat{J}_\alpha$ are already Hermitian by (4.27), and the electron sector is flat. Conjugating the identity part of (4.31), working from the innermost derivative outward (each line is one product rule):

$$\begin{aligned} \partial_{Q_b}\big(\gamma^{-1/4}\psi_W\big) &= \gamma^{-1/4}\Big[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\, \psi_W\Big], \\ \gamma^{1/2}\, \partial_{Q_b}\big(\gamma^{-1/4}\psi_W\big) &= \gamma^{1/4}\Big[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\, \psi_W\Big], \\ \partial_{Q_b}\Big(\gamma^{1/4}\big[\cdots\big]\Big) &= \gamma^{1/4}\Big[\partial^2_{Q_b}\psi_W - \tfrac{1}{16}(\partial_{Q_b}\ln\gamma)^2\, \psi_W - \tfrac{1}{4}(\partial^2_{Q_b}\ln\gamma)\, \psi_W\Big], \end{aligned} \tag{4.33}$$

where in the last line the two cross terms $\pm\frac{1}{4}(\partial_{Q_b}\ln\gamma)\, \partial_{Q_b}\psi_W$ (one from differentiating $\gamma^{1/4}$, one from the bracket) cancel. Multiplying by $-\frac{\hbar^2}{2}\gamma^{1/4}\cdot\gamma^{-1/2} = -\frac{\hbar^2}{2}\gamma^{-1/4}$ and summing over b: the identity part becomes the bare $\frac{1}{2}\sum_b \hat{P}_b^2$ plus a pure multiplicative function,

$$U_{\text{vib}} = \frac{\hbar^2}{8}\sum_b \Big[\partial_{Q_b}^2 \ln\gamma + \tfrac{1}{4}\big(\partial_{Q_b}\ln\gamma\big)^2\Big]. \tag{4.34}$$

The Coriolis block responds even more simply. Conjugating (4.30): in the first group, $\gamma^{1/4}\hat{P}_b\gamma^{-1/4} = \hat{P}_b + \frac{i\hbar}{4}(\partial_{Q_b}\ln\gamma)$; in the second, $\gamma^{-1/4}\hat{P}_b\gamma^{1/4} = \hat{P}_b - \frac{i\hbar}{4}(\partial_{Q_b}\ln\gamma)$. The two induced terms are equal and opposite [they multiply the same function, since $\hat{J}_\alpha\mu_{\alpha\beta}\sigma_{\beta b} = \sigma_{\beta b}\mu_{\alpha\beta}\hat{J}_\alpha$ upon relabeling the summed Cartesian indices — these are functions of different variables and commute], so they cancel, leaving the manifestly symmetric flat-measure form

$$\hat{T}_{\text{Cor},W} = -\tfrac{1}{2}\sum_{\alpha\beta}\hat{J}_\alpha\big(\mu_{\alpha\beta}\,\hat{\pi}_\beta + \hat{\pi}_\beta\,\mu_{\alpha\beta}\big) \;=\; -\tfrac{1}{2}\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big), \tag{4.35}$$

with *no* extra potential from this block. Finally, conjugating the $\boldsymbol{\sigma}^T\boldsymbol{\mu}\boldsymbol{\sigma}$ dressing of (4.31) and reorganizing it into the symmetric kernel ordering $\frac{1}{2}\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta$ leaves a second multiplicative residue, U_{Cor} (every derivative of this bookkeeping is written out in Appendix B).

The total extra potential $\hat{U} = U_{\text{vib}} + U_{\text{Cor}}$ collapses through two exact identities, both pure Q-calculus. First, Jacobi's formula — for a symmetric positive-definite matrix, $\ln\det\mathbf{I}' = \operatorname{tr}\ln\mathbf{I}'$, so differentiating,

$$\partial_{Q_b}\ln\gamma = \operatorname{tr}\!\big(\boldsymbol{\mu}\,\partial_{Q_b}\mathbf{I}'\big), \qquad \partial_{Q_b}\boldsymbol{\mu} = -\,\boldsymbol{\mu}\,(\partial_{Q_b}\mathbf{I}')\,\boldsymbol{\mu} \quad [\text{from } \partial_{Q_b}(\boldsymbol{\mu}\,\mathbf{I}') = 0], \tag{4.36}$$

expresses every derivative of γ and $\boldsymbol{\mu}$ through $\partial_{Q_b}\mathbf{I}'$. Second, the Eckart conditions endow $\mathbf{I}'$ with a remarkable exact closed form, due to Watson: with the *constant* equilibrium inertia derivatives $a^b \equiv (\partial\mathbf{I}_N/\partial Q_b)|_{Q=0}$,

$$\mathbf{I}'(Q) = A(Q)\,(\mathbf{I}^e)^{-1}A(Q), \qquad A(Q) \equiv \mathbf{I}^e + \tfrac{1}{2}\sum_b a^b\,Q_b, \qquad \mathbf{I}^e \equiv \mathbf{I}_N(0), \tag{4.37}$$

proven term by term from the Eckart completeness relations in Appendix B (§ B.6). Hence $\gamma = (\det A)^2/\det\mathbf{I}^e$ and $\boldsymbol{\mu} = A^{-1}\mathbf{I}^e A^{-1}$: every derivative of γ and $\boldsymbol{\mu}$ reduces to the constant matrices a^b. Substituting (4.36)–(4.37) into $U_{\text{vib}} + U_{\text{Cor}}$ and collapsing the sums with the Eckart sum rules — the complete term-by-term algebra is § B.6 — the outcome, first obtained by Watson [3], is the single trace

$$\boxed{\hat{U}(Q) \;=\; -\frac{\hbar^2}{8}\sum_{\alpha=1}^{3}\mu_{\alpha\alpha}(Q), \qquad \boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^\mathsf{T})^{-1}.} \tag{4.38}$$

Two remarks. *(a)* The collapse is special to *rectilinear* normal coordinates: for general curvilinear internal coordinates the closed form (4.37) fails and additional extra-potential terms survive. *(b)* The electrons do not modify $\hat{U}$: γ and $\boldsymbol{\mu}$ depend on Q alone, and the electronic Jacobian is flat (§ 4.4). The pseudopotential is a c-number on the nuclear shape space, fixed entirely by the modified inertia $\mathbf{I}'$.

4.10 Electronic kinetic operator

In BF Cartesian electron coordinates, $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on $L^2(\mathbb{R}^{3N_{\rm elec}}, \prod_i d^3\bar{\mathbf{r}}_i)$ (§ 4.4). The electronic + mass-polarization kinetic operator is therefore the straightforward quantization of (3.21):

$$\hat{T}_e \;=\; \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2. \tag{4.39}$$

Expanding the second piece:

$$\frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 \;=\; \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2M_N} \;+\; \frac{1}{M_N}\sum_{i<l} \hat{\bar{\mathbf{p}}}_i\cdot\hat{\bar{\mathbf{p}}}_l. \tag{4.40}$$

The diagonal $1/(2M_N)$ piece renormalizes the effective electron mass through $1/m_e + 1/M_N = M_{\rm tot}/(m_e M_N) = 1/\mu_e$. The off-diagonal two-electron term is the mass-polarization *sensu stricto*, dynamically coupling electrons through nuclear recoil — a purely classical effect (Section 2.4) inherited by the quantum theory.

4.11 Complete quantum molecular Hamiltonian

Collect the flat-measure blocks: the rotational kernel (4.29), the Coriolis coupling (4.35), the vibrational operator [bare $\frac{1}{2}\sum_b \hat{P}_b^2$ plus the symmetric $\frac{1}{2}\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta$], the pseudopotential (4.38), the electronic operator (4.39), the shift $\hat{J}_\alpha \to \hat{J}_\alpha - \hat{L}_{{\rm elec},\alpha}$ of § 4.4, and the rotationally invariant Coulomb potentials of (3.8). The pieces assemble into the sandwich-ordered kernel [the term-by-term identification of the nine products with the blocks above, including every derivative, is Appendix B]:

$$\boxed{\begin{aligned}\hat{H}_{\rm int} \;=\; &\tfrac{1}{2}\sum_{\alpha\beta}\big(\hat{J}_\alpha - \hat{L}_{{\rm elec},\alpha} - \hat{\pi}_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat{J}_\beta - \hat{L}_{{\rm elec},\beta} - \hat{\pi}_\beta\big)\\ &+\tfrac{1}{2}\sum_b \hat{P}_b^2\\ &+\sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2\\ &-\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}(Q)\\ &+V_{NN}(Q) \;+\; V_{Ne}(Q,\{\bar{\mathbf{r}}_i\}) \;+\; V_{ee}(\{\bar{\mathbf{r}}_i\}),\end{aligned}} \tag{4.41}$$

acting on $L^2(\mathcal{C},\, d\mu_W)$ with $d\mu_W = \sin\theta\, d\phi\, d\theta\, d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$. Every operator in (4.41) is an explicit differential operator in the coordinates — the $\hat{J}_\alpha$ are the Euler-angle combinations (4.24), $\hat{\pi}_\alpha = \sum_b \sigma_{\alpha b}\hat{P}_b$, $\hat{P}_b = -i\hbar\partial_{Q_b}$, $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ — and every one is Hermitian on $d\mu_W$ by the integration-by-parts arguments given above.

This is the **exact non-relativistic molecular Hamiltonian** after separation of the total COM motion, in BF + Eckart frame, with electrons referred to the nuclear COM. The only approximations made along the way were: (i) non-relativistic dynamics; (ii) Eckart's choice of the BF orientation (a convention, not an approximation); (iii) parametrization of the nuclear shape by $3N_{\rm nucl} - 6$ internal coordinates Q_b (a parametrization, also exact).

Remark (the anomalous commutator, not needed). The traditional route to (4.41) [14, 7, 4] defines the body-fixed angular momentum by projecting the space-fixed operator through the direction cosines, derives its *anomalous* commutation relations $[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\,\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$, and uses that algebra to control the operator ordering of the rotational kernel. Nothing in this section used it: the $\hat{J}_\alpha$ entered only as the shorthand (4.24) for explicit Euler-angle derivatives, Hermiticity came from integration by parts with the divergence identity (4.26), and the pseudopotential came from the Q-derivatives of the volume element. The anomalous algebra is a correct and sometimes useful property of the operators (4.24), but it is a *consequence* of the coordinate representation, not an input to the quantization.

4.12 Born–Huang expansion

4.12.1 Clamped-nuclei electronic Hamiltonian

At each nuclear configuration Q, define the **BF clamped-nuclei electronic Hamiltonian** as the sum of all Q-parametric electron-only pieces of (4.41):

$$\hat{H}_{\rm el}(Q) \;\equiv\; \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^{\,2}}{2m_e} \;+\; \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 \;+\; V_{Ne}(Q,\{\bar{\mathbf{r}}_i\}) \;+\; V_{ee}(\{\bar{\mathbf{r}}_i\}). \tag{4.42}$$

Remark. The mass-polarization piece $(\sum_i \hat{\bar{\mathbf{p}}}_i)^2/(2M_N)$ is included here as a small correction with $1/M_N$. Some treatments ("clamped-nuclei BO") drop it; we keep it explicit because it is parametric in Q only through the absent dependence (i.e., M_N is just a number), which means it acts trivially on the parametric family of electronic Hilbert spaces.

Solve the parametric eigenvalue problem:

$$\hat{H}_{\rm el}(Q)\,|\phi_n(Q)\rangle \;=\; E_n^{\rm el}(Q)\,|\phi_n(Q)\rangle, \tag{4.43}$$

in the electronic Hilbert space $\mathcal{H}_e = L^2(\mathbb{R}^{3N_{\rm elec}}, d^{3N_{\rm elec}}\bar{\mathbf{r}})$. The eigenstates $\{|\phi_n(Q)\rangle\}$ depend *parametrically* on Q — Q is treated as a fixed classical parameter, not a quantum degree of freedom — and form a complete orthonormal basis of $\mathcal{H}_e$ at each Q:

$$\sum_n |\phi_n(Q)\rangle\langle\phi_n(Q)| \;=\; \hat{\mathbb{1}}_e, \qquad \langle\phi_m(Q)|\phi_n(Q)\rangle_e = \delta_{mn}. \tag{4.44}$$

4.12.2 The exact Born–Huang expansion

The *exact* molecular wavefunction $|\Psi\rangle \in L^2(\mathcal{C}, d\mu_W)$ admits a resolution against the parametric electronic basis. Insert the resolution of identity (4.44) at each Q:

$$|\Psi\rangle \;=\; \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle, \qquad \chi_n(\Omega, Q) \;\equiv\; \langle\phi_n(Q)|\Psi\rangle_e, \tag{4.45}$$

where $\Omega = (\phi, \theta, \chi)$, the electronic inner product is taken at fixed Q, and χ_n are the **nuclear coefficient functions**. The expansion (4.45) is **exact**, not an ansatz: it follows from the completeness (4.44) at each Q, with the only subtlety being that the basis $|\phi_n(Q)\rangle$ varies parametrically. There is *no* simple tensor factorization $\mathcal{H}_{\rm molecule} = \mathcal{H}_N \otimes \mathcal{H}_e$ because of this parametric dependence; the Born–Huang expansion [11] is the rigorous replacement.

4.12.3 Coupled-channel equations

Substitute (4.45) into $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$, project from the left by $\langle\phi_m(Q)|$, and use (4.43):

$$\big[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)\big]\chi_n(\Omega, Q) \; + \; \sum_m \hat{\Lambda}_{nm}\,\chi_m(\Omega, Q) \; = \; E\,\chi_n(\Omega, Q), \quad (4.46)$$

where $\hat{T}_N$ denotes the nuclear (rotational + vibrational) kinetic operator from the first two lines of (4.41). The **non-adiabatic coupling operators** $\hat{\Lambda}_{nm}$ arise from the action of $\hat{T}_N$ (and the rot–electronic couplings through $\hat{\mathbf{L}}_{\text{elec}}$) on the Q-parametric dependence of $|\phi_m(Q)\rangle$:

$$\hat{\Lambda}_{nm} \;\sim\; -\hbar^2\,\langle\phi_n|\nabla_Q\phi_m\rangle_e\cdot\nabla_Q \;-\; \frac{\hbar^2}{2}\,\langle\phi_n|\nabla_Q^2\phi_m\rangle_e \;+\; \ldots, \quad (4.47)$$

with analogous terms from the rotation–electron Coriolis $\hat{J}_\alpha\hat{L}_{\text{elec},\alpha}$ kernel.

4.12.4 Born–Oppenheimer approximation as truncation

The **Born–Oppenheimer (BO) approximation** is the diagonal truncation of (4.46): set $\hat{\Lambda}_{nm} \to 0$ for $m \neq n$ and (optionally) also $\hat{\Lambda}_{nn} \to 0$. The resulting effective nuclear problem is

$$\big[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)\big]\chi_n^{\text{BO}}(\Omega, Q) \; = \; E^{\text{BO}}\,\chi_n^{\text{BO}}(\Omega, Q), \quad (4.48)$$

where the **Born–Oppenheimer potential energy surface** for electronic state n is

$$V_n^{\text{BO}}(Q) \;\equiv\; E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \quad (4.49)$$

The exactness of (4.45) and the well-definedness of (4.46) are mathematical facts. BO is the physical assumption that off-diagonal $\hat{\Lambda}_{nm}$ are small — justified when electronic gaps are large compared to nuclear kinetic energies, breaking down at conical intersections and avoided crossings, where the parametric variation of $|\phi_n(Q)\rangle$ becomes singular.

4.12.5 Summary

1. (4.41) is the exact non-relativistic quantum molecular Hamiltonian in BF + Eckart + nuclear-COM coordinates, obtained entirely from coordinate derivatives: metric (4.15) $\to$ measure (4.21) $\to$ inverse (4.22) $\to$ Laplace–Beltrami blocks (4.29)–(4.31) $\to$ rescaling and pseudopotential (4.38).

2. The Born–Huang expansion (4.45) is exact, following from completeness of the parametric electronic basis (4.44).

3. The infinite coupled system (4.46) is equivalent to the Schrödinger equation $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$.

4. The Born–Oppenheimer truncation reduces it to (4.48), giving the familiar single-PES picture (4.49).

Appendix

Appendix A: Mass-Polarization Term in Classical Molecular Hamiltonian

Section 2 produced the mass-polarization term on the momentum side, via the generating function (2.5). This appendix re-derives it from the velocity side: substitute the same mixed mass-centered coordinates of § 2.1 into the laboratory-frame kinetic energy, read off the canonical momenta from the Lagrangian, and Legendre-transform back. Every step is elementary; the payoff is seeing where the mass polarization hides in velocity form — as a *negative* correction — and how the Legendre transform flips its sign.

We start from the laboratory-frame kinetic energy

$$T = \frac{1}{2}\sum_j M_j \dot{\mathbf{R}}_j^2 + \frac{1}{2}\sum_i m_e \dot{\mathbf{r}}_i^2, \tag{1}$$

where M_j is the mass of the j-th nucleus, $\mathbf{R}_j$ its Cartesian coordinate, m_e the electron mass, and $\mathbf{r}_i$ the coordinate of the i-th electron.

Mixed Mass-Centered Coordinates

As in (2.1)–(2.3): the nuclear and total masses and centers of mass are

$$M_N = \sum_j M_j, \qquad M_{\text{tot}} = M_N + N_{\text{elec}}\, m_e, \tag{2}$$

$$\mathbf{R}_{\text{ncm}} = \frac{1}{M_N}\sum_j M_j \mathbf{R}_j, \qquad \mathbf{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}}\Big(\sum_j M_j \mathbf{R}_j + m_e \sum_i \mathbf{r}_i\Big) = \mathbf{R}_{\text{ncm}} + \frac{m_e}{M_{\text{tot}}}\sum_i \mathbf{r}'_i, \tag{3}$$

and the internal coordinates are referred to the *nuclear* CoM,

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{ncm}}, \qquad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{ncm}}, \qquad \sum_j M_j \mathbf{R}'_j = 0. \tag{4}$$

The constraint ties the nuclear relative vectors only; the $\mathbf{r}'_i$ are unconstrained. To substitute into T we need the old coordinates in terms of the new. Solving the second equality of the $\mathbf{R}_{\text{cm}}$ definition for the nuclear CoM,

$$\mathbf{R}_{\text{ncm}} = \mathbf{R}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}}\sum_l \mathbf{r}'_l, \tag{5}$$

and therefore

$$\mathbf{R}_j = \mathbf{R}_{\text{cm}} + \mathbf{R}'_j - \frac{m_e}{M_{\text{tot}}}\sum_l \mathbf{r}'_l, \qquad \mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}'_i - \frac{m_e}{M_{\text{tot}}}\sum_l \mathbf{r}'_l. \tag{6}$$

Both families ride on the *same* drift — the velocity of the nuclear CoM,

$$\mathbf{D} \equiv \dot{\mathbf{R}}_{\text{ncm}} = \dot{\mathbf{R}}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}}\sum_l \dot{\mathbf{r}}'_l, \qquad \dot{\mathbf{R}}_j = \mathbf{D} + \dot{\mathbf{R}}'_j, \qquad \dot{\mathbf{r}}_i = \mathbf{D} + \dot{\mathbf{r}}'_i. \tag{7}$$

Kinetic Energy in Velocity Form

Substitute the velocities into T. The nuclear part, using $\sum_j M_j\dot{\mathbf{R}}'_j = 0$ (the time derivative of the constraint) on the cross term,

$$T_N = \frac{1}{2}\sum_j M_j\big(\mathbf{D}+\dot{\mathbf{R}}'_j\big)^2 = \frac{1}{2}M_N\mathbf{D}^2 + \mathbf{D}\cdot\underbrace{\sum_j M_j\dot{\mathbf{R}}'_j}_{=0} + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j. \tag{8}$$

The electronic part has no constraint, so its cross term survives:

$$T_e = \frac{1}{2}m_e\sum_i\big(\mathbf{D}+\dot{\mathbf{r}}'_i\big)^2 = \frac{1}{2}N_{\text{elec}}\,m_e\mathbf{D}^2 + m_e\,\mathbf{D}\cdot\sum_i\dot{\mathbf{r}}'_i + \frac{1}{2}m_e\sum_i\dot{\mathbf{r}}'^2_i. \tag{9}$$

Adding, with $M_N + N_{\text{elec}}m_e = M_{\text{tot}}$,

$$T = \frac{1}{2}M_{\text{tot}}\mathbf{D}^2 + m_e\,\mathbf{D}\cdot\sum_i\dot{\mathbf{r}}'_i + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j + \frac{1}{2}m_e\sum_i\dot{\mathbf{r}}'^2_i. \tag{10}$$

Now expand $\mathbf{D} = \dot{\mathbf{R}}_{\text{cm}} - \frac{m_e}{M_{\text{tot}}}\sum_l\dot{\mathbf{r}}'_l$ in the first two terms:

$$\frac{1}{2}M_{\text{tot}}\mathbf{D}^2 = \frac{1}{2}M_{\text{tot}}\dot{\mathbf{R}}^2_{\text{cm}} - m_e\,\dot{\mathbf{R}}_{\text{cm}}\cdot\sum_l\dot{\mathbf{r}}'_l + \frac{m_e^2}{2M_{\text{tot}}}\Big(\sum_l\dot{\mathbf{r}}'_l\Big)^2, \tag{11}$$

$$m_e\,\mathbf{D}\cdot\sum_i\dot{\mathbf{r}}'_i = m_e\,\dot{\mathbf{R}}_{\text{cm}}\cdot\sum_i\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\text{tot}}}\Big(\sum_l\dot{\mathbf{r}}'_l\Big)^2. \tag{12}$$

The two $\dot{\mathbf{R}}_{\text{cm}}\cdot\sum\dot{\mathbf{r}}'$ cross terms cancel *identically* — the velocity-form image of the momentum-side cancellation in § 2.3 — and the quadratic pieces combine to a single negative remainder:

$$\boxed{T = \frac{1}{2}M_{\text{tot}}\dot{\mathbf{R}}^2_{\text{cm}} + \frac{1}{2}\sum_j M_j\dot{\mathbf{R}}'^2_j + \frac{1}{2}m_e\sum_i\dot{\mathbf{r}}'^2_i - \frac{m_e^2}{2M_{\text{tot}}}\Big(\sum_i\dot{\mathbf{r}}'_i\Big)^2.} \tag{13}$$

Two features are already visible at the velocity level: the CoM decouples with the *correct* total mass M_{tot}, and the electron sector carries a *negative* collective correction. That negative term is the mass polarization in disguise — the Legendre transform below flips its sign and trades M_{tot} for M_N.

Canonical Momenta

With $L = T - V$ and V velocity-independent, differentiate the boxed T.
CoM: only the first term carries $\dot{\mathbf{R}}_{\text{cm}}$, so

$$\boxed{\mathbf{P}_{\text{cm}} = \frac{\partial L}{\partial\dot{\mathbf{R}}_{\text{cm}}} = M_{\text{tot}}\dot{\mathbf{R}}_{\text{cm}}.} \tag{14}$$

Sanity check: $\sum_j M_j\dot{\mathbf{R}}_j + m_e\sum_i\dot{\mathbf{r}}_i = M_{\text{tot}}\mathbf{D} + m_e\sum_i\dot{\mathbf{r}}'_i = M_{\text{tot}}\dot{\mathbf{R}}_{\text{cm}}$, so $\mathbf{P}_{\text{cm}}$ is the total momentum, as it must be.

Electrons: the last two terms of the boxed T carry $\dot{\mathbf{r}}'_i$,

$$\boxed{\mathbf{p}'_i = \frac{\partial L}{\partial \dot{\mathbf{r}}'_i} = m_e\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\rm tot}}\sum_l \dot{\mathbf{r}}'_l,} \tag{15}$$

which is precisely the momentum–velocity relation (3.5). Summing over i,

$$\sum_i \mathbf{p}'_i = m_e\Big(1 - \frac{N_{\rm elec}m_e}{M_{\rm tot}}\Big)\sum_l \dot{\mathbf{r}}'_l = \frac{m_e M_N}{M_{\rm tot}}\sum_l \dot{\mathbf{r}}'_l, \tag{16}$$

so $\frac{m_e^2}{M_{\rm tot}}\sum_l \dot{\mathbf{r}}'_l = \frac{m_e}{M_N}\sum_l \mathbf{p}'_l$, and the relation inverts to

$$m_e\dot{\mathbf{r}}'_i = \mathbf{p}'_i + \frac{m_e}{M_N}\sum_l \mathbf{p}'_l \qquad \Longleftrightarrow \qquad \dot{\mathbf{r}}'_i = \frac{\mathbf{p}'_i}{m_e} + \frac{1}{M_N}\sum_l \mathbf{p}'_l, \tag{17}$$

the Hamilton equation (3.4).

Nuclei: on the constraint surface $\sum_j M_j\dot{\mathbf{R}}'_j = 0$ and in the dual gauge $\sum_j \mathbf{P}'_j = 0$ of § 2.2,

$$\boxed{\mathbf{P}'_j = M_j\dot{\mathbf{R}}'_j.} \tag{18}$$

Consistency check against (2.8): the lab momentum is $M_j\dot{\mathbf{R}}_j = M_j\mathbf{D} + M_j\dot{\mathbf{R}}'_j$, and

$$\mathbf{D} = \dot{\mathbf{R}}_{\rm cm} - \frac{m_e}{M_{\rm tot}}\sum_l \dot{\mathbf{r}}'_l = \frac{\mathbf{P}_{\rm cm}}{M_{\rm tot}} - \frac{1}{M_N}\sum_l \mathbf{p}'_l, \tag{19}$$

which is exactly the common vector $\mathbf{b}$ of (2.9); hence $\mathbf{P}_j = \frac{M_j}{M_{\rm tot}}\mathbf{P}_{\rm cm} + \mathbf{P}'_j - \frac{M_j}{M_N}\sum_l \mathbf{p}'_l$, reproducing (2.8).

Kinetic Energy in Momentum Form

The CoM and nuclear terms are immediate: $\frac{1}{2}M_{\rm tot}\dot{\mathbf{R}}_{\rm cm}^2 = \mathbf{P}_{\rm cm}^2/2M_{\rm tot}$ and $\frac{1}{2}M_j\dot{\mathbf{R}}_j'^2 = \mathbf{P}_j'^2/2M_j$. For the electron sector, abbreviate $\bar{\mathbf{P}} \equiv \sum_l \mathbf{p}'_l$ and substitute $\dot{\mathbf{r}}'_i = \mathbf{p}'_i/m_e + \bar{\mathbf{P}}/M_N$:

$$\frac{1}{2}m_e\sum_i \dot{\mathbf{r}}_i'^2 = \sum_i \frac{1}{2m_e}\Big(\mathbf{p}'_i + \frac{m_e}{M_N}\bar{\mathbf{P}}\Big)^2 = \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\bar{\mathbf{P}}^2}{M_N} + \frac{N_{\rm elec}m_e}{2M_N^2}\bar{\mathbf{P}}^2, \tag{20}$$

and, using $\sum_l \dot{\mathbf{r}}'_l = \frac{M_{\rm tot}}{m_e M_N}\bar{\mathbf{P}}$,

$$-\frac{m_e^2}{2M_{\rm tot}}\Big(\sum_l \dot{\mathbf{r}}'_l\Big)^2 = -\frac{m_e^2}{2M_{\rm tot}}\cdot\frac{M_{\rm tot}^2}{m_e^2M_N^2}\,\bar{\mathbf{P}}^2 = -\frac{M_{\rm tot}}{2M_N^2}\,\bar{\mathbf{P}}^2. \tag{21}$$

Collecting the three $\bar{\mathbf{P}}^2$ coefficients,

$$\frac{1}{M_N} + \frac{N_{\rm elec}m_e - M_{\rm tot}}{2M_N^2} = \frac{1}{M_N} - \frac{M_N}{2M_N^2} = \frac{1}{M_N} - \frac{1}{2M_N} = +\frac{1}{2M_N}, \tag{22}$$

using $M_{\rm tot} - N_{\rm elec} m_e = M_N$. The negative velocity-form correction has become the *positive* mass polarization:

$$\boxed{T = \frac{\mathbf{P}_{\rm cm}^2}{2M_{\rm tot}} + \sum_j \frac{\mathbf{P}_j^{'2}}{2M_j} + \sum_i \frac{\mathbf{p}_i^{'2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2.} \tag{23}$$

Hamiltonian

The potential is translationally invariant and both internal families share the reference point $\mathbf{R}_{\rm ncm}$, so every interparticle separation is a plain coordinate difference and $V = V(\{\mathbf{R}'_j\}, \{\mathbf{r}'_i\})$. Hence

$$\boxed{H = \frac{\mathbf{P}_{\rm cm}^2}{2M_{\rm tot}} + \sum_j \frac{\mathbf{P}_j^{'2}}{2M_j} + \sum_i \frac{\mathbf{p}_i^{'2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2 + V(\{\mathbf{R}'_j\}, \{\mathbf{r}'_i\}),} \tag{24}$$

identical to (2.12) — exact for *any* $\mathbf{P}_{\rm cm}$, with no rest-frame choice, no CoM–internal cross-coupling, and the correct total mass in the free term.

Important Interpretation

The canonical electron momentum is not $m_e\dot{\mathbf{r}}'_i$. It is depleted by the collective recoil,

$$\mathbf{p}'_i = m_e\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\rm tot}} \sum_l \dot{\mathbf{r}}'_l, \tag{25}$$

because moving one electron drags the total CoM, and the coordinate $\mathbf{r}'_i$ is measured from the (recoiling) nuclear CoM.
Why the sign flips. The electron block of the velocity-form kinetic energy is the quadratic form $\frac{1}{2}\dot{\mathbf{r}}'^{\mathsf{T}}\mathbf{A}\,\dot{\mathbf{r}}'$ with the non-diagonal mass matrix

$$\mathbf{A} = m_e\Big(\mathbb{1} - \frac{m_e}{M_{\rm tot}}\mathbf{J}\Big), \qquad \mathbf{J}_{il} = 1 \text{ (all-ones block)}, \tag{26}$$

and the Legendre transform replaces $\mathbf{A}$ by its *inverse*:

$$\mathbf{A}^{-1} = \frac{1}{m_e}\Big(\mathbb{1} + \frac{m_e}{M_N}\mathbf{J}\Big) \qquad \Longrightarrow \qquad \frac{1}{2}\,\mathbf{p}'^{\mathsf{T}}\mathbf{A}^{-1}\mathbf{p}' = \sum_i \frac{\mathbf{p}_i^{'2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2, \tag{27}$$

(one verifies $\mathbf{A}\mathbf{A}^{-1} = \mathbb{1}$ using $\mathbf{J}^2 = N_{\rm elec}\mathbf{J}$ and $M_{\rm tot} - N_{\rm elec}m_e = M_N$). The inversion turns the negative $-m_e^2/2M_{\rm tot}$ collective coupling into the positive $+1/2M_N$ mass polarization: same physics, opposite side of the Legendre transform.
Bookkeeping. The accompanying constraints are

$$\boxed{\sum_j M_j\mathbf{R}'_j = 0, \qquad \sum_j \mathbf{P}'_j = 0,} \tag{28}$$

purely nuclear on both sides. Everything above agrees with the generating-function route of Section 2: the momentum relations reproduce (2.8), and the Hamiltonian reproduces (2.12).

Appendix B: Term-by-term expansion of the Podolsky operator, with every derivative

Section 4 quantized the classical Hamiltonian (3.22) by applying the Podolsky operator (4.5) to the ro-vibrational metric (4.15) and rescaling to the flat measure. This appendix supplies what the main text compressed: the expansion of the quantized kernel of (4.41),

$$\hat{T}_{\text{kernel}} = \tfrac{1}{2}\sum_{\alpha\beta} \hat{A}_\alpha\, \mu_{\alpha\beta}(Q)\, \hat{A}_\beta, \qquad \hat{A}_\alpha \equiv \hat{J}_\alpha - \hat{L}_\alpha - \hat{\pi}_\alpha, \qquad \hat{L}_\alpha \equiv \hat{L}_{\text{elec},\alpha}, \tag{B.1}$$

into its nine products, with *every* coordinate derivative written out and the Hermiticity of each block verified from first principles. All arguments use only: the adjoint of a product reverses the order, $(\hat{X}\hat{Y})^\dagger = \hat{Y}^\dagger\hat{X}^\dagger$; integration by parts on the measure $d\mu_W$; and the product rule. No commutator algebra of body-fixed angular momenta appears anywhere (cf. the Remark in § 4.11). Repeated Cartesian indices $\alpha, \beta \in \{1,2,3\}$ and mode indices $b, c \in \{1,\ldots,f\}$ are summed.

B.1 Toolkit

The three momentum-like operators live in three separate coordinate sectors:

$$\hat{J}_\alpha = -i\hbar\,(\mathsf{E}^{-1})_{s\alpha}\partial_{\Omega_s}\ \text{[eq. (4.24), Euler angles]}, \quad \hat{P}_b = -i\hbar\,\partial_{Q_b}\ \text{[modes]}, \quad \hat{L}_\alpha = -i\hbar\,\epsilon_{\alpha\beta\gamma}\textstyle\sum_i \bar{r}_{i\beta}\partial_{\bar{r}_{i\gamma}}\ \text{[electrons]}. \tag{B.2}$$

Because they differentiate disjoint sets of variables, *operators from different sectors commute*, and each sector's functions are constants for the other sectors' derivatives. Each is Hermitian on its factor of $d\mu_W$: $\hat{J}_\alpha$ on $\sin\theta\, d\Omega$ by the divergence identity (4.26) [via (4.27)], $\hat{L}_\alpha$ on the flat electron measure by (4.18), and $\hat{P}_b$ on the flat $\prod_b dQ_b$ trivially. The only non-commutativity in the problem is the product rule within the Q-sector,

$$\hat{P}_b\, f(Q) = f(Q)\,\hat{P}_b - i\hbar\,(\partial_{Q_b} f) \qquad \Longrightarrow \qquad \hat{P}_b\,\mu_{\alpha\beta} = \mu_{\alpha\beta}\hat{P}_b - i\hbar\,\partial_b\mu_{\alpha\beta}, \quad \hat{P}_b\,\sigma_{\alpha c} = \sigma_{\alpha c}\hat{P}_b - i\hbar\,\zeta^\alpha_{bc}, \tag{B.3}$$

where the last relation uses $\sigma_{\alpha b} = \sum_a \zeta^\alpha_{ab} Q_a$ [(3.17a)], linear in Q with *constant* $\zeta^\alpha_{ab} = \left(\sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}\right)_\alpha$, so $\partial_{Q_b}\sigma_{\alpha c} = \zeta^\alpha_{bc}$. The antisymmetry of the cross product gives the sum rule that recurs throughout:

$$\zeta^\alpha_{ab} = -\zeta^\alpha_{ba} \qquad \Longrightarrow \qquad \zeta^\alpha_{bb} = 0, \qquad \sum_b \partial_{Q_b}\sigma_{\alpha b} = \sum_b \zeta^\alpha_{bb} = 0. \tag{B.4}$$

$\hat{\pi}_\alpha$ is Hermitian on the flat mode measure. With $\hat{\pi}_\alpha = \sum_b \sigma_{\alpha b}\hat{P}_b$, take the adjoint (order reverses; σ real, $\hat{P}_b$ Hermitian), then slide $\hat{P}_b$ back to the right with (B.3), then apply (B.4):

$$\hat{\pi}_\alpha^\dagger = \sum_b \hat{P}_b\,\sigma_{\alpha b} = \sum_b \left(\sigma_{\alpha b}\hat{P}_b - i\hbar\,\zeta^\alpha_{bb}\right) = \hat{\pi}_\alpha. \tag{B.5}$$

Had $\sum_b \zeta^\alpha_{bb}$ been nonzero, $\hat{\pi}_\alpha$ would have required explicit symmetrization; the antisymmetry of ζ spares us. Finally, $\mu_{\alpha\beta}(Q) = \mu_{\beta\alpha}(Q)$ is a real symmetric function of Q alone.

B.2 The nine products

Substitute $\hat{A}_\alpha = \hat{J}_\alpha - \hat{L}_\alpha - \hat{\pi}_\alpha$ into (B.1) and multiply out, *keeping the operator order* (the factors need not commute):

$$\begin{aligned}\hat{A}_\alpha\mu_{\alpha\beta}\hat{A}_\beta = \; & \underbrace{\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta}_{\text{(T1) rotation}} + \underbrace{\hat{L}_\alpha\mu_{\alpha\beta}\hat{L}_\beta}_{\text{(T2) electronic}} + \underbrace{\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta}_{\text{(T3) vibrational}} - \underbrace{\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{L}_\beta + \hat{L}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big)}_{\text{(T4) rotation–electron}} \\ & - \underbrace{\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big)}_{\text{(T5) rotation–vibration}} + \underbrace{\big(\hat{L}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha\mu_{\alpha\beta}\hat{L}_\beta\big)}_{\text{(T6) vibration–electron}}. \end{aligned} \tag{B.6}$$

Each cross term collects the two equal classical contributions $X\mu Y$ and $Y\mu X$, which is why it appears as a symmetric sum of orderings. We now treat the six blocks one at a time: for each we verify Hermiticity and write the explicit differential form.

B.3 Diagonal terms

(T1) $\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta$**.** *Hermiticity.* Take the adjoint (μ real, $\hat{J}_\alpha^\dagger = \hat{J}_\alpha$ on $\sin\theta\, d\Omega$; order reverses), then relabel the summed dummy indices $\alpha \leftrightarrow \beta$, then use $\mu_{\beta\alpha} = \mu_{\alpha\beta}$:

$$\big(\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta\big)^\dagger = \hat{J}_\beta\,\mu_{\alpha\beta}\,\hat{J}_\alpha = \hat{J}_\alpha\,\mu_{\beta\alpha}\,\hat{J}_\beta = \hat{J}_\alpha\,\mu_{\alpha\beta}\,\hat{J}_\beta. \tag{B.7}$$

Hermitian exactly as written — no symmetrization, no commutator evaluated. *Differential form.* $\mu(Q)$ commutes with the Euler derivatives, and inserting (B.2),

$$\hat{J}_\alpha\mu_{\alpha\beta}\hat{J}_\beta = -\hbar^2\mu_{\alpha\beta}\,(\mathsf{E}^{-1})_{s\alpha}\Big[(\mathsf{E}^{-1})_{t\beta}\,\partial_{\Omega_s}\partial_{\Omega_t} + \big(\partial_{\Omega_s}(\mathsf{E}^{-1})_{t\beta}\big)\,\partial_{\Omega_t}\Big], \tag{B.8}$$

with the trigonometric entries of E^{-1} read off (4.12). Every term carries at least one Euler derivative: the rotational block generates *no* pure multiplicative term, in agreement with the Laplace–Beltrami computation (4.29).

(T2) $\hat{L}_\alpha\mu_{\alpha\beta}\hat{L}_\beta$**.** Word-for-word the same as (T1) with $\hat{J} \to \hat{L}$: the adjoint chain (B.7) uses only $\hat{L}_\alpha^\dagger = \hat{L}_\alpha$ [(4.18)] and the symmetry of μ. Hermitian as written; all terms carry electron derivatives; no multiplicative residue.

(T3) $\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta$**.** *Hermiticity.* Use $\hat{\pi}^\dagger = \hat{\pi}$ from (B.5), then relabel $\alpha \leftrightarrow \beta$ and use μ symmetric — the same three-step chain as (B.7). Hermitian as written, but *only* because $\hat{\pi}$ is Hermitian, i.e. ultimately because of the antisymmetry (B.4). *Differential form.* Write $\hat{\pi}_\alpha = \sigma_{\alpha b}\hat{P}_b$ and slide the left momentum past the Q-function $\mu_{\alpha\beta}\sigma_{\beta c}$ with (B.3):

$$\begin{aligned}\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta &= \sigma_{\alpha b}\,\hat{P}_b\,\big(\mu_{\alpha\beta}\sigma_{\beta c}\big)\,\hat{P}_c = \sigma_{\alpha b}\Big[\big(\mu_{\alpha\beta}\sigma_{\beta c}\big)\hat{P}_b - i\hbar\,\partial_b\big(\mu_{\alpha\beta}\sigma_{\beta c}\big)\Big]\hat{P}_c \\ &= \underbrace{G_{bc}\,\hat{P}_b\hat{P}_c}_{\text{2nd order}} - i\hbar\,\underbrace{\sigma_{\alpha b}\Big[(\partial_b\mu_{\alpha\beta})\sigma_{\beta c} + \mu_{\alpha\beta}\,\zeta^\beta_{bc}\Big]\hat{P}_c}_{\text{1st order}}, \qquad G_{bc} \equiv (\boldsymbol{\sigma}^\mathsf{T}\boldsymbol{\mu}\boldsymbol{\sigma})_{bc}, \end{aligned} \tag{B.9}$$

the Coriolis-dressed vibrational operator. Every surviving term carries at least one $\hat{P}$: again no pure multiplicative term. *Equivalence with the ordered form.* The same sum rule shows that (T3) equals the fully left-ordered operator: moving the left $\hat{P}_b$ past its own coefficient

$\sigma_{\alpha b}$,

$$\hat P_b\, G_{bc}\, \hat P_c - \hat\pi_\alpha \mu_{\alpha\beta}\hat\pi_\beta = \big[\hat P_b, \sigma_{\alpha b}\big]\, \mu_{\alpha\beta}\sigma_{\beta c}\hat P_c = -i\hbar\Big(\textstyle\sum_b \zeta^{\alpha}_{bb}\Big)\mu_{\alpha\beta}\sigma_{\beta c}\hat P_c = 0, \tag{B.10}$$

so $\frac{1}{2}\hat\pi_\alpha\mu_{\alpha\beta}\hat\pi_\beta = \frac{1}{2}\hat P_b\, G_{bc}\, \hat P_c$ *exactly.* This identity is what lets the sandwich-ordered kernel of (4.41) match the Laplace–Beltrami vibrational block (4.31), whose dressed part is precisely $\frac{1}{2}\hat P_b G_{bc}\hat P_c$ after the rescaling (see § B.5).

B.4 Cross terms

(T4) $\hat J_\alpha\mu_{\alpha\beta}\hat L_\beta + \hat L_\alpha\mu_{\alpha\beta}\hat J_\beta$**.** All three factors act on disjoint variables and mutually commute. A single ordering is already Hermitian,

$$\big(\hat J_\alpha\mu_{\alpha\beta}\hat L_\beta\big)^\dagger = \hat L_\beta\, \mu_{\alpha\beta}\, \hat J_\alpha = \mu_{\alpha\beta}\, \hat J_\alpha \hat L_\beta = \hat J_\alpha\mu_{\alpha\beta}\hat L_\beta, \tag{B.11}$$

and the two orderings in the sum are equal (relabel $\alpha \leftrightarrow \beta$ and commute), so

$$\hat J_\alpha\mu_{\alpha\beta}\hat L_\beta + \hat L_\alpha\mu_{\alpha\beta}\hat J_\beta = 2\,\mu_{\alpha\beta}\,\hat J_\alpha\hat L_\beta \qquad \text{(Hermitian; no residue).} \tag{B.12}$$

This is the rotation–electron Coriolis coupling; no $\hat P$ appears, so there is no differential subtlety.

(T5) $\hat J_\alpha\mu_{\alpha\beta}\hat\pi_\beta + \hat\pi_\alpha\mu_{\alpha\beta}\hat J_\beta$**.** Now the subtlety: $\hat\pi$ does *not* commute with μ, since by (B.3) $\hat\pi_\beta\mu_{\alpha\beta} = \mu_{\alpha\beta}\hat\pi_\beta - i\hbar\,\sigma_{\beta b}\partial_b\mu_{\alpha\beta} \neq \mu_{\alpha\beta}\hat\pi_\beta$. A single ordering is therefore *not* Hermitian by itself; its adjoint is exactly the *other* ordering ($\hat J$ commutes with μ and $\hat\pi$; relabel $\alpha \leftrightarrow \beta$):

$$\big(\hat J_\alpha\mu_{\alpha\beta}\hat\pi_\beta\big)^\dagger = \hat\pi_\beta\, \mu_{\alpha\beta}\, \hat J_\alpha = \hat\pi_\alpha\, \mu_{\alpha\beta}\, \hat J_\beta. \tag{B.13}$$

The cross term is of the form $\hat X + \hat X^\dagger$, automatically Hermitian — *this is why the classical product* $-2J\mu\pi$ *must be quantized as the symmetric sum*, and it is precisely the symmetric sum that the Laplace–Beltrami operator produced by itself in (4.30)/(4.35). Pulling $\hat J$ out front and expanding the anticommutator with (B.3),

$$\hat J_\alpha\mu_{\alpha\beta}\hat\pi_\beta + \hat\pi_\alpha\mu_{\alpha\beta}\hat J_\beta = \hat J_\alpha\big(\mu_{\alpha\beta}\hat\pi_\beta + \hat\pi_\beta\mu_{\alpha\beta}\big) = \hat J_\alpha\Big(\underbrace{2\,\mu_{\alpha\beta}\sigma_{\beta c}\hat P_c}_{\text{1st order in } \hat P}\ \underbrace{-\,i\hbar\,\sigma_{\beta c}\,\partial_c\mu_{\alpha\beta}}_{\text{0th order in } \hat P}\Big). \tag{B.14}$$

The first piece is the rotation–vibration Coriolis coupling ($\hat J$ times $\hat P$); the second is the imaginary, $\hat J$-linear counterterm that makes the sum Hermitian. It is *not* a stand-alone multiplicative residue — it still carries $\hat J_\alpha$ — consistent with the Laplace–Beltrami result (4.35) that the Coriolis block contributes nothing to the pseudopotential. As a cross-check of (B.14) against the pre-rescaling form (4.30): expanding the second group of (4.30) by the product rule,

$$\tfrac{1}{\sqrt{\gamma}}\,\hat P_b\,\sqrt{\gamma}\,\sigma_{\alpha b}\mu_{\alpha\beta}\,\hat J_\beta = \Big[\sigma_{\alpha b}\mu_{\alpha\beta}\hat P_b - i\hbar\,\partial_b\big(\sigma_{\alpha b}\mu_{\alpha\beta}\big) - \tfrac{i\hbar}{2}\,(\partial_b \ln\gamma)\,\sigma_{\alpha b}\mu_{\alpha\beta}\Big]\hat J_\beta, \tag{B.15}$$

and conjugating by $\gamma^{1/4}$ as in § 4.9 cancels the $\partial_b \ln\gamma$ terms between the two groups, reproducing exactly $-\frac{1}{2}\hat J_\alpha\{\mu_{\alpha\beta}, \hat\pi_\beta\}$ of (4.35), i.e. $-\frac{1}{2}\times$(B.14).

(T6) $\hat{L}_\alpha \mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta}\hat{L}_\beta$**.** Identical to (T5) with $\hat{J} \to \hat{L}$ ($\hat{L}$ also commutes with μ and $\hat{\pi}$):

$$\hat{L}_\alpha \mu_{\alpha\beta}\hat{\pi}_\beta + \hat{\pi}_\alpha \mu_{\alpha\beta}\hat{L}_\beta = \hat{L}_\alpha \Big(2\,\mu_{\alpha\beta}\sigma_{\beta c}\hat{P}_c - i\hbar\,\sigma_{\beta c}\,\partial_c \mu_{\alpha\beta}\Big), \tag{B.16}$$

the vibration–electron angular coupling.

B.5 The measure residue U_{Cor}, every derivative

Section 4.9 derived the identity-block residue U_{vib} [(4.34)] in full and deferred the residue of the dressed block. Here it is. Conjugate the dressed part of the vibrational Laplace–Beltrami block (4.31) by $\gamma^{1/4}$ and act on ψ_W; each line is one product rule [$G_{bc} = (\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\mu}\boldsymbol{\sigma})_{bc}$, symmetric]:

$$\begin{aligned}
\partial_{Q_c}\big(\gamma^{-1/4}\psi_W\big) &= \gamma^{-1/4}\Big[\partial_{Q_c}\psi_W - \tfrac{1}{4}(\partial_c \ln\gamma)\psi_W\Big],\\
\gamma^{1/2}\,G_{bc}\,\partial_{Q_c}\big(\gamma^{-1/4}\psi_W\big) &= \gamma^{1/4}\,G_{bc}\Big[\partial_{Q_c}\psi_W - \tfrac{1}{4}(\partial_c \ln\gamma)\psi_W\Big],\\
\partial_{Q_b}\Big(\gamma^{1/4}\,G_{bc}\big[\cdots\big]\Big) &= \gamma^{1/4}\Big[G_{bc}\,\partial_b\partial_c\psi_W + (\partial_b G_{bc})\,\partial_c\psi_W + \tfrac{1}{4}(\partial_b \ln\gamma)G_{bc}\,\partial_c\psi_W - \tfrac{1}{4}G_{bc}(\partial_c\ln\gamma)\,\partial_b\psi_W\\
&\qquad - \tfrac{1}{4}\,\partial_b\big(G_{bc}\,\partial_c\ln\gamma\big)\psi_W - \tfrac{1}{16}\,G_{bc}(\partial_b\ln\gamma)(\partial_c\ln\gamma)\,\psi_W\Big].
\end{aligned} \tag{B.17}$$

The two first-order $\ln\gamma$ terms cancel by the symmetry of G_{bc} (relabel $b \leftrightarrow c$), leaving $G\partial\partial + (\partial G)\partial$ — which is exactly $-\frac{2}{\hbar^2} \times \frac{1}{2}\hat{P}_b G_{bc}\hat{P}_c$ — plus a pure multiplicative remainder. Multiplying by $-\frac{\hbar^2}{2}$ (the prefactors $\gamma^{1/4}\cdot\gamma^{-1/2}\cdot\gamma^{1/4} = 1$ combine to unity),

$$\gamma^{1/4}\,\hat{T}_{\mathrm{dress}}\,\gamma^{-1/4} = \tfrac{1}{2}\,\hat{P}_b\,G_{bc}\,\hat{P}_c + U_{\mathrm{Cor}}, \qquad \boxed{U_{\mathrm{Cor}} = \frac{\hbar^2}{8}\Big[\partial_b\big(G_{bc}\,\partial_c\ln\gamma\big) + \tfrac{1}{4}\,G_{bc}\,(\partial_b\ln\gamma)(\partial_c\ln\gamma)\Big].} \tag{B.18}$$

By (B.10), $\frac{1}{2}\hat{P}_b G_{bc}\hat{P}_c = \frac{1}{2}\hat{\pi}_\alpha\mu_{\alpha\beta}\hat{\pi}_\beta$, so the operator part is exactly the (T3) sandwich of the kernel (4.41), and the entire ordering debt of the dressed block is the multiplicative U_{Cor}. Together with U_{vib} of (4.34) [the same computation with $G_{bc} \to \delta_{bc}$], the total residue is

$$\hat{U} = U_{\mathrm{vib}} + U_{\mathrm{Cor}} = \frac{\hbar^2}{8}\Big[\partial_b\big(g^{QQ}_{bc}\,\partial_c\ln\gamma\big) + \tfrac{1}{4}\,g^{QQ}_{bc}\,(\partial_b\ln\gamma)(\partial_c\ln\gamma)\Big], \qquad g^{QQ}_{bc} = \delta_{bc} + G_{bc}, \tag{B.19}$$

the complete term-by-term collapse of (B.19) to the trace $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$ of (4.38) [3] — via Jacobi's formula (4.36), Watson's closed form (4.37), and the Eckart sum rules — is carried out in § B.6.

The division of labour is clean. In the explicit forms (B.8), (B.9), (B.12), (B.14), (B.16), every surviving kernel term carries at least one derivative operator ($\hat{P}$, $\hat{J}$, or $\hat{L}$): *the ordering of the kernel never generates a pure multiplicative term.* The pseudopotential (4.38) is an $\mathcal{O}(\hbar^2)$ c-number that originates entirely in the *curved integration measure* $\sqrt{\gamma}$, extracted by the rescaling $\psi_W = \gamma^{1/4}\psi$ — (B.18) is its dressed-block share. Kernel ordering → the Hermitian operators of this appendix; the measure → the pseudopotential.

B.6 From (B.19) to the Watson trace (4.38): every step

Equation (B.19) expresses the total residue through two shape functions — $\gamma = \det\mathbf{I}'$ and the Coriolis dressing $G_{bc} = (\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\mu}\boldsymbol{\sigma})_{bc}$ — and its collapse to the single trace (4.38) rests entirely on the *Eckart sum rules*: completeness relations obeyed by the mode vectors $\mathbf{l}_{jb}$. This subsection derives each of them and then assembles (B.19) term by term. Throughout, $\Lambda_b \equiv \partial_b \ln\gamma$ with $\partial_b \equiv \partial_{Q_b}$, repeated indices are summed, and all matrices are 3×3 unless stated.

Step 0: expansion into five groups. In $\sum_b \partial_b(G_{bc}\Lambda_c)$ write $G_{bc}\Lambda_c = \sigma_{\alpha b}(\boldsymbol{\mu}\mathbf{v})_\alpha$ with the 3-vector $v_\alpha \equiv \sum_c \sigma_{\alpha c}\Lambda_c$; the derivative of the explicit $\sigma_{\alpha b}$ dies by (B.4), $\sum_b \partial_b\sigma_{\alpha b} = \sum_b \zeta^\alpha_{bb} = 0$. The product rule then splits (B.19) into five groups,

$$\frac{8}{\hbar^2}\,\hat U = \underbrace{\sum_b\Big[\partial_b\Lambda_b + \tfrac14\Lambda_b^2\Big]}_{N_0} + \underbrace{\sigma_{\alpha b}(\partial_b\boldsymbol{\mu})_{\alpha\beta}\,v_\beta}_{N_1} + \underbrace{\sigma_{\alpha b}\,\mu_{\alpha\beta}\,\zeta^\beta_{bc}\,\Lambda_c}_{N_2} + \underbrace{G_{bc}\,\partial_b\Lambda_c}_{N_3} + \underbrace{\tfrac14\,\mathbf{v}^{\mathsf T}\boldsymbol{\mu}\,\mathbf{v}}_{N_4}, \qquad \text{(B.20)}$$

where $N_1+N_2+N_3$ came from $\sigma_{\alpha b}\partial_b(\boldsymbol{\mu}\mathbf{v})_\alpha$ via $\partial_b v_\beta = \zeta^\beta_{bc}\Lambda_c + \sigma_{\beta c}\,\partial_b\Lambda_c$. We now build the tools, evaluate each group, and add them up.

Step 1: the Eckart constants. Write $\mathbf{e}_j \equiv \sqrt{M_j}\,\bar{\mathbf{R}}^{\rm eq}_j$ and $\mathbf{u}_j(Q) \equiv \sqrt{M_j}\,\bar{\mathbf{R}}_j(Q) = \mathbf{e}_j + \sum_b \mathbf{l}_{jb}Q_b$, so that $\mathbf{I}_N = \sum_j(u_j^2\mathbb{1} - \mathbf{u}_j\mathbf{u}_j^{\mathsf T})$ and $\sigma_{\alpha b} = \sum_j(\mathbf{u}_j\times\mathbf{l}_{jb})_\alpha$. Define the constants

$$S_{\alpha\beta} \equiv \sum_j e_{j\alpha}e_{j\beta}, \qquad \mathbf{I}^e = (\operatorname{tr} S)\,\mathbb{1} - S, \qquad \boldsymbol{\mu}^e \equiv (\mathbf{I}^e)^{-1}, \qquad a^b_{\alpha\beta} \equiv \sum_j\big[2(\mathbf{e}_j{\cdot}\mathbf{l}_{jb})\delta_{\alpha\beta} - l_{jb,\alpha}e_{j\beta} - e_{j\alpha}l_{jb,\beta}\big], \qquad \text{(B.21)}$$

the a^b being the equilibrium inertia derivatives, $a^b = \partial\mathbf{I}_N/\partial Q_b|_{Q=0}$. The rotational Eckart condition $\sum_j \mathbf{e}_j\times\mathbf{l}_{jb} = 0$ says precisely that $E^b_{\alpha\beta} \equiv \sum_j e_{j\alpha}l_{jb,\beta}$ is *symmetric*; comparing with (B.21) and taking a trace,

$$E^b = s_b\,\mathbb{1} - \tfrac12\,a^b, \qquad s_b \equiv \sum_j \mathbf{e}_j\cdot\mathbf{l}_{jb} = \tfrac14\operatorname{tr} a^b. \qquad \text{(B.22)}$$

Define $A(Q) \equiv \mathbf{I}^e + \frac12\sum_b a^bQ_b$ (symmetric, constant derivatives $\partial_c A = \frac12 a^c$). Then (B.22) gives two exact corollaries used throughout [for the second: $\sum_b Q_bE^b = (\sum_b s_bQ_b)\mathbb{1} - (A-\mathbf{I}^e)$ with $\sum_b s_bQ_b = \frac14\operatorname{tr}\sum_b Q_ba^b = \frac12\operatorname{tr}(A-\mathbf{I}^e)$, and $\operatorname{tr}\mathbf{I}^e = 2\operatorname{tr} S$]:

$$\sum_b Q_b\,a^b = 2\,(A-\mathbf{I}^e), \qquad Y_{\alpha\beta} \equiv \sum_j u_{j\alpha}e_{j\beta} = S_{\alpha\beta} + \sum_b Q_bE^b_{\alpha\beta} = \tfrac12(\operatorname{tr} A)\,\delta_{\alpha\beta} - A_{\alpha\beta}. \qquad \text{(B.23)}$$

Step 2: completeness and the two workhorse contractions. The $3N_{\rm nucl}$ mass-weighted displacement vectors — the f vibrations $\mathbf{l}_b$, three translations $\propto \sqrt{M_j}\,\hat{\mathbf{n}}_\alpha$, and three rotations $t^\gamma_{j\beta} = (\hat{\mathbf{n}}_\gamma\times\mathbf{e}_j)_\beta$ (mutually orthogonal by the Eckart conditions; rotational Gram matrix $\sum_j \mathbf{t}^\gamma_j\cdot\mathbf{t}^\delta_j = I^e_{\gamma\delta}$) — form a complete set. Resolving the identity on $\mathbb{R}^{3N_{\rm nucl}}$,

$$\sum_b l_{jb,\alpha}\,l_{kb,\beta} = \delta_{jk}\delta_{\alpha\beta} - \frac{\sqrt{M_jM_k}}{M_N}\,\delta_{\alpha\beta} - \epsilon_{\gamma\rho\alpha}\,e_{j\rho}\,\mu^e_{\gamma\delta}\,\epsilon_{\delta\sigma\beta}\,e_{k\sigma}. \qquad \text{(B.24)}$$

Every contraction below pairs (B.24) with coefficients built from $\mathbf{e}_j$, $\mathbf{u}_j$, or $\mathbf{l}_{jc}$, all orthogonal to the translations ($\sum_j\sqrt{M_j}\,\mathbf{e}_j = \sum_j\sqrt{M_j}\,\mathbf{u}_j = \sum_j\sqrt{M_j}\,\mathbf{l}_{jc} = 0$), so *the translational* *term never contributes.* The rotational term always meets one of two ϵ-contractions, evaluated once and for all with $\epsilon_{\alpha\gamma\delta}\epsilon_{\gamma''\rho\delta} = \delta_{\alpha\gamma''}\delta_{\gamma\rho} - \delta_{\alpha\rho}\delta_{\gamma\gamma''}$, (B.22), and (B.23):

$$\epsilon_{\alpha\gamma\delta}\,\epsilon_{\gamma''\rho\delta}\sum_j l_{jc,\gamma}\,e_{j\rho} = s_c\,\delta_{\alpha\gamma''} - E^c_{\alpha\gamma''} = \tfrac{1}{2}\,a^c_{\alpha\gamma''}, \qquad \epsilon_{\alpha\gamma\delta}\,\epsilon_{\gamma''\rho\delta}\sum_j u_{j\gamma}\,e_{j\rho} = \tfrac{1}{2}(\operatorname{tr}A)\,\delta_{\alpha\gamma''} - Y_{\gamma''\alpha} = A_{\alpha\gamma''}. \tag{B.25}$$

Step 3: the four sum rules. Insert (B.24) into each vibrational-index contraction and evaluate the rotational corrections with (B.25). For the $\zeta\zeta$ contraction, $\zeta^\alpha_{bd}\zeta^\beta_{cd} = \epsilon_{\alpha\gamma\delta}\epsilon_{\beta\varepsilon\varphi}\sum_{jk} l_{jb,\gamma}l_{kc,\varepsilon}\,[\sum_d l_{jd,\delta}l_{kd,\varphi}]$: the δ_{jk} part contracts the two ϵ's directly, the rotational part produces $\frac{1}{2}a^b$ on each side, giving

$$\sum_d \zeta^\alpha_{bd}\,\zeta^\beta_{cd} = \delta_{\alpha\beta}\,\delta_{bc} - \sum_j l_{jb,\beta}\,l_{jc,\alpha} - \tfrac{1}{4}\big(a^b\boldsymbol{\mu}^e a^c\big)_{\alpha\beta}. \tag{B.26}$$

By the same pattern (the $\mathbf{u}$-side of a $\boldsymbol{\sigma}$ produces A, an $\mathbf{l}$- or a-side produces $\frac{1}{2}a$), with the shorthand

$$V[M]^\alpha_{\gamma\delta} \equiv \epsilon_{\alpha\rho\gamma}M_{\rho\delta} + \epsilon_{\alpha\rho\delta}M_{\rho\gamma} \quad (\text{any symmetric } M;\ V[\mathbb{1}] = 0), \tag{B.27}$$

one finds the aa, σa, and $\sigma\zeta$ rules

$$\begin{aligned}\sum_b a^b_{yz}\,a^b_{wx} &= 4(\operatorname{tr}S)\delta_{yz}\delta_{wx} - 4S_{wx}\delta_{yz} - 4S_{yz}\delta_{wx} + \delta_{yw}S_{zx} + \delta_{yx}S_{zw} + \delta_{zw}S_{yx} + \delta_{zx}S_{yw}\\ &\quad - V[S]^\gamma_{yz}\,\mu^e_{\gamma\delta}\,V[S]^\delta_{wx}, \qquad (B.28)\\ \sum_b \sigma_{\alpha b}\,a^b_{\gamma\delta} &= V[A]^\alpha_{\gamma\delta} + (A\boldsymbol{\mu}^e)_{\alpha\varepsilon}\,V[S]^\varepsilon_{\gamma\delta}, \qquad (B.29)\\ \sum_b \sigma_{\alpha b}\,\zeta^\beta_{bc} &= \sum_j u_{j\beta}\,l_{jc,\alpha} - \delta_{\alpha\beta}\,(s_c + Q_c) + \tfrac{1}{2}\big(A\boldsymbol{\mu}^e a^c\big)_{\alpha\beta}. \qquad (B.30)\end{aligned}$$

Step 4: Watson's closed form for $\mathbf{I}'$ [the identity (4.37)]. Expand $\mathbf{I}_N$ and $\boldsymbol{\sigma}\boldsymbol{\sigma}^\mathsf{T}$ exactly (both are quadratic polynomials in Q; the linear coefficient of $\mathbf{I}_N$ is a^b by (B.22)):

$$\mathbf{I}_{N,\alpha\beta} = \mathbf{I}^e_{\alpha\beta} + \sum_b a^b_{\alpha\beta}Q_b + \sum_{b,c} Q_bQ_c\Big[\delta_{bc}\delta_{\alpha\beta} - \sum_j l_{jb,\alpha}l_{jc,\beta}\Big], \qquad (\boldsymbol{\sigma}\boldsymbol{\sigma}^\mathsf{T})_{\alpha\beta} = \sum_{b,c} Q_bQ_c\sum_d \zeta^\alpha_{bd}\zeta^\beta_{cd}. \tag{B.31}$$

Subtracting and inserting (B.26), the δ-terms cancel, the two $\sum_j l\,l$ terms cancel under the $b \leftrightarrow c$ symmetrization enforced by Q_bQ_c, and only the $\frac{1}{4}a^b\boldsymbol{\mu}^e a^c$ term survives — which is exactly the quadratic term of $A\boldsymbol{\mu}^e A$:

$$\boxed{\mathbf{I}'(Q) = A(Q)\,\boldsymbol{\mu}^e A(Q), \qquad A(Q) = \mathbf{I}^e + \tfrac{1}{2}\sum_b a^b Q_b,} \tag{B.32}$$

the closed form quoted as (4.37). Its corollaries do all the differential work below:

$$\boldsymbol{\mu} = A^{-1}\mathbf{I}^e A^{-1}, \qquad \boldsymbol{\mu}A = A^{-1}\mathbf{I}^e, \qquad A\boldsymbol{\mu}A = \mathbf{I}^e, \qquad \boldsymbol{\mu}A\boldsymbol{\mu}^e = A^{-1}, \qquad \gamma = \frac{(\det A)^2}{\det \mathbf{I}^e}, \tag{B.33}$$

and hence, with $B^b \equiv A^{-1}a^b$ and Jacobi's formula (4.36) applied to $\det A$,

$$\Lambda_b = \partial_b \ln\gamma = 2\,\mathrm{tr}\big(A^{-1}\partial_b A\big) = \mathrm{tr}\,B^b, \qquad \partial_c\Lambda_b = -\tfrac{1}{2}\,\mathrm{tr}\big(B^bB^c\big): \tag{B.34}$$

every derivative of γ and $\boldsymbol{\mu}$ reduces to the *constant* matrices a^b.

Step 5: the identity group N_0. By (B.34), $N_0 = \sum_b[-\frac{1}{2}\,\mathrm{tr}(B^bB^b) + \frac{1}{4}(\mathrm{tr}\,B^b)^2]$. Both sums are contractions of (B.28) with two factors of A^{-1}; carrying them out term by term (each Kronecker term becomes a trace) and writing $\kappa_\gamma \equiv \epsilon_{\gamma\sigma\tau}(SA^{-1})_{\sigma\tau} = \frac{1}{2}\,\mathrm{tr}(A^{-1}V[S]^\gamma)$ for the antisymmetric part of SA^{-1}:

$$\begin{aligned} N_0 =\; &\mathrm{tr}\,S\,\big[(\mathrm{tr}\,A^{-1})^2 - 2\,\mathrm{tr}\,A^{-2}\big] + 4\,\mathrm{tr}(A^{-2}S) - 3\,\mathrm{tr}(A^{-1})\,\mathrm{tr}(A^{-1}S) \\ &+ \tfrac{1}{2}\,\mu^e_{\gamma\delta}\,\mathrm{tr}\big(A^{-1}V[S]^\gamma A^{-1}V[S]^\delta\big) - \boldsymbol{\kappa}^\mathsf{T}\boldsymbol{\mu}^e\boldsymbol{\kappa}. \end{aligned} \tag{B.35}$$

Step 6: N_1 vanishes (a divergence lemma). From (B.33), $\partial_b\boldsymbol{\mu} = -\frac{1}{2}(B^b\boldsymbol{\mu}+\boldsymbol{\mu}B^{b\mathsf{T}})$, so N_1 needs $\sum_b \sigma_{\alpha b}B^b_{\alpha q}$ and $\sum_b \sigma_{\alpha b}(\boldsymbol{\mu}B^{b\mathsf{T}})_{\alpha r}$, both fixed by (B.29). Each vanishes identically: $A^{-1}_{\alpha p}V[A]^\alpha_{pq} = 0$ and $\mu_{\alpha p}V[A]^\alpha_{pq}$-type contractions reduce, via $A^{-1}A = \mathbb{1}$, $\boldsymbol{\mu}A\boldsymbol{\mu}^e = A^{-1}$, and the commutativity of S with μ^e (both are functions of S), to ϵ-contractions of *symmetric* matrices, which are zero, plus a pair $\pm\kappa_r$ that cancels. Hence

$$\sum_b \sigma_{\alpha b}\,(\partial_b\boldsymbol{\mu})_{\alpha\beta} = 0 \qquad\Longleftrightarrow\qquad \sum_b \partial_b\,(\boldsymbol{\mu}\boldsymbol{\sigma})_{\alpha b} = 0, \qquad \text{so} \qquad N_1 = 0: \tag{B.36}$$

the three vector fields $(\boldsymbol{\mu}\boldsymbol{\sigma})_{\alpha b}$ are divergence-free on the mode space.

Step 7: the quadratic group N_4. Contracting (B.29) with A^{-1} gives $v_\alpha = \sum_b \sigma_{\alpha b}\,\mathrm{tr}\,B^b$: the $V[A]$ part dies [$\mathrm{tr}(A^{-1}V[A]^\alpha) = 0$ by the same symmetric-ϵ argument], leaving $v_\alpha = 2(A\boldsymbol{\mu}^e\boldsymbol{\kappa})_\alpha$. Then, by $A\boldsymbol{\mu}A = \mathbf{I}^e$,

$$N_4 = \tfrac{1}{4}\,\mathbf{v}^\mathsf{T}\boldsymbol{\mu}\,\mathbf{v} = \boldsymbol{\kappa}^\mathsf{T}\boldsymbol{\mu}^e\,(A\boldsymbol{\mu}A)\,\boldsymbol{\mu}^e\boldsymbol{\kappa} = \boldsymbol{\kappa}^\mathsf{T}\boldsymbol{\mu}^e\boldsymbol{\kappa}, \tag{B.37}$$

which cancels the last term of (B.35) on the spot.

Step 8: the Coriolis-dressed group N_3. By (B.34), $N_3 = -\frac{1}{2}\,G_{bc}\,\mathrm{tr}(B^bB^c) = -\frac{1}{2}\,\mu_{\alpha\beta}\,\mathrm{tr}\big(A^{-1}W_\alpha A^{-1}W_\beta\big)$ with $W_\alpha \equiv \sum_b \sigma_{\alpha b}a^b$ from (B.29). Expanding $W = V[A] + A\boldsymbol{\mu}^eV[S]$ and using $\boldsymbol{\mu}(A\boldsymbol{\mu}^e) = A^{-1}$ and $(A\boldsymbol{\mu}^e)^\mathsf{T}\boldsymbol{\mu}(A\boldsymbol{\mu}^e) = \boldsymbol{\mu}^e$:

$$N_3 = -\tfrac{1}{2}\,\mu_{\alpha\beta}\,\mathrm{tr}\big(A^{-1}V[A]^\alpha A^{-1}V[A]^\beta\big) - A^{-1}_{\alpha\varepsilon}\,\mathrm{tr}\big(A^{-1}V[A]^\alpha A^{-1}V[S]^\varepsilon\big) - \tfrac{1}{2}\,\mu^e_{\varepsilon\varphi}\,\mathrm{tr}\big(A^{-1}V[S]^\varepsilon A^{-1}V[S]^\varphi\big), \tag{B.38}$$

whose last term cancels the μ^e-quadratic term of (B.35). All $\boldsymbol{\mu}^e$-dependence is now gone from the running total.

Step 9: the mixed group N_2. Contract (B.30) with $\mu_{\alpha\beta}\Lambda_c$. Three auxiliary sums close the two non-constant pieces. First, contracting (B.24) once more [with the coefficients of s_c and of Λ_c; the rotational corrections vanish by symmetry of S],

$$\sum_c s_c\,a^c = 2\,\mathbf{I}^e \quad\Longrightarrow\quad \sum_c (s_c + Q_c)\Lambda_c = \mathrm{tr}\Big(A^{-1}\big[2\mathbf{I}^e + 2(A - \mathbf{I}^e)\big]\Big) = 6. \tag{B.39}$$

Second, the same completeness argument applied to $\ell_{j\alpha} \equiv \sum_c l_{jc,\alpha}\Lambda_c$ and to $P \equiv \sum_c \Lambda_c\, a^c$ [a contraction of (B.28)] gives

$$\ell_{j\alpha} = 2\big[(\operatorname{tr} A^{-1})\, e_{j\alpha} - (A^{-1}\mathbf{e}_j)_\alpha\big] + 2\,\epsilon_{\gamma\rho\alpha}\, e_{j\rho}\, (\boldsymbol{\mu}^e\boldsymbol{\kappa})_\gamma\,, \qquad \tfrac{1}{2}\operatorname{tr}\big(A^{-1}P\big) = 2\operatorname{tr} S(\operatorname{tr} A^{-1})^2 - 4\operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S) + 2\operatorname{tr}(A^{-2}S \tag{B.40}$$

Assembling [$\sum_j u_{j\beta} l_{jc,\alpha}$ contracts to $\sum_j(\boldsymbol{\mu}\mathbf{u}_j)\cdot\boldsymbol{\ell}_j$, evaluated with $Y = \frac{1}{2}(\operatorname{tr} A)\mathbb{1} - A$ of (B.23) and $\operatorname{tr}(A^{-1}\boldsymbol{\mu}A) = \operatorname{tr}\boldsymbol{\mu}$; the term $-\delta_{\alpha\beta}(s_c + Q_c)$ contributes $-6\operatorname{tr}\boldsymbol{\mu}$ by (B.39); the last term of (B.30) contributes $\frac{1}{2}\operatorname{tr}(A^{-1}P)$ via $\boldsymbol{\mu}A\boldsymbol{\mu}^e = A^{-1}$; the two $\boldsymbol{\kappa}^\mathsf{T}\boldsymbol{\mu}^e\boldsymbol{\kappa}$ terms cancel between $\boldsymbol{\ell}_j$ and P]:

$$N_2 = \operatorname{tr} A\left[\operatorname{tr} A^{-1}\operatorname{tr}\boldsymbol{\mu} - \operatorname{tr}(A^{-1}\boldsymbol{\mu})\right] - 2\operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S) - 4\operatorname{tr}\boldsymbol{\mu} + 2\operatorname{tr}(A^{-2}S). \tag{B.41}$$

Step 10: the two remaining ϵ-traces. The V-traces left in (B.38) are evaluated with the pair identity (any symmetric M)

$$M_{ij}\,\epsilon_{iab}\,\epsilon_{jcd} = \operatorname{tr} M(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}) - M_{ac}\delta_{bd} + M_{bc}\delta_{ad} + M_{ad}\delta_{bc} - M_{bd}\delta_{ac}, \tag{B.42}$$

together with $A^{-1}A = \mathbb{1}$ (which collapses every $V[A]$ factor). The results are

$$\mu_{\alpha\beta}\operatorname{tr}\big(A^{-1}V[A]^\alpha A^{-1}V[A]^\beta\big) = 2\operatorname{tr} A\left[\operatorname{tr} A^{-1}\operatorname{tr}\boldsymbol{\mu} - \operatorname{tr}(A^{-1}\boldsymbol{\mu})\right] - 6\operatorname{tr}\boldsymbol{\mu} - 2\operatorname{tr} A^{-1}\big[\operatorname{tr} S\operatorname{tr} A^{-1} - \operatorname{tr}(A^{-1}S)\big], \tag{B.43}$$

$$A^{-1}_{\alpha\varepsilon}\operatorname{tr}\big(A^{-1}V[A]^\alpha A^{-1}V[S]^\varepsilon\big) = 2\operatorname{tr} S\left[(\operatorname{tr} A^{-1})^2 - \operatorname{tr} A^{-2}\right] - 6\operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S) + 6\operatorname{tr}(A^{-2}S). \tag{B.44}$$

Step 11: assembly. Add the five groups. The $\boldsymbol{\kappa}^\mathsf{T}\boldsymbol{\mu}^e\boldsymbol{\kappa}$ terms cancelled between (B.35) and (B.37); the μ^e-quadratic $V[S]$ traces cancelled between (B.35) and (B.38); the $\operatorname{tr} A[\cdots]$ terms cancel between (B.41) and the first line of (B.43). What remains is a short list of ordinary traces:

$$\begin{aligned}
\frac{8}{\hbar^2}\hat{U} = {} & \underbrace{\operatorname{tr} S\big[(\operatorname{tr} A^{-1})^2 - 2\operatorname{tr} A^{-2}\big] + 4\operatorname{tr}(A^{-2}S) - 3\operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S)}_{\text{from } N_0}\underbrace{- 2\operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S) - 4\operatorname{tr}\boldsymbol{\mu} + 2\operatorname{tr}(A^{-2}S)}_{\text{from } N_2} \\
& \underbrace{+ 3\operatorname{tr}\boldsymbol{\mu} + \operatorname{tr} S(\operatorname{tr} A^{-1})^2 - \operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S)}_{\text{from } -\frac{1}{2}\times\text{(B.43)}}\underbrace{- 2\operatorname{tr} S\big[(\operatorname{tr} A^{-1})^2 - \operatorname{tr} A^{-2}\big] + 6\operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S) - 6\operatorname{tr}(A^{-2}S)}_{\text{from } -\text{(B.44)}} \\
= {} & -\operatorname{tr}\boldsymbol{\mu} \qquad \big[\text{all } (\operatorname{tr} A^{-1})^2,\ \operatorname{tr} A^{-2},\ \operatorname{tr}(A^{-2}S),\ \operatorname{tr} A^{-1}\operatorname{tr}(A^{-1}S) \text{ coefficients vanish; } \operatorname{tr}\boldsymbol{\mu} = \operatorname{tr} S\operatorname{tr} A^{-2} - \operatorname{tr}(A^{-2}
\end{aligned} \tag{B.45}$$

that is,

$$\boxed{\hat{U}(Q) = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}(Q)} \tag{B.46}$$

— Watson's trace formula (4.38), obtained from (B.19) using only the product rule, Jacobi's formula, and the Eckart completeness relation (B.24). No commutator of body-fixed angular momenta was used at any step.

B.7 Worked example: a bent triatomic

To make "every derivative" concrete, take a bent triatomic XY_2 (e.g. H_2O), a C_{2v} asymmetric top with $N_{\text{nucl}} = 3$ and $f = 3N_{\text{nucl}} - 6 = 3$ normal coordinates: the symmetric stretch Q_1 and bend Q_2 (both a_1) and the antisymmetric stretch Q_3 (b_2). Choose the body frame as the instantaneous principal-axis system, with a, b in the molecular plane and $c \perp$ to it.

Inertia and its derivatives. At equilibrium the inertia tensor is diagonal, $\mathbf{I}_N^{\text{eq}} = \text{diag}(I_a^e, I_b^e, I_c^e)$, and because the equilibrium mass distribution is planar the perpendicular-axis theorem gives the exact relation

$$I_c^e = I_a^e + I_b^e. \tag{B.47}$$

The totally symmetric coordinates Q_1, Q_2 preserve the C_{2v} symmetry, so to linear order they only *scale* the diagonal moments, whereas the antisymmetric Q_3 tilts the principal axes and enters $\mathbf{I}_N$ only at second order. Hence

$$I_\alpha(Q) = I_\alpha^e + a_\alpha^1 Q_1 + a_\alpha^2 Q_2 + \mathcal{O}(Q^2), \qquad a_\alpha^b \equiv \frac{\partial I_\alpha}{\partial Q_b}\bigg|_{\text{eq}}, \tag{B.48}$$

with the inertial derivatives a_α^b computable from the mode vectors $\mathbf{l}_{jb}$ evaluated at equilibrium. These are exactly the derivatives that populate the rotation–vibration Jacobian $\sqrt{\gamma(Q)}$ of (4.21).

Coriolis coupling. The only nonzero constant Coriolis matrix in C_{2v} couples the two a_1 modes to the b_2 mode about the b axis (the axis whose rotation maps a symmetric into an antisymmetric displacement):

$$\boldsymbol{\sigma}_1 = \zeta_{13}^b\, Q_3\, \hat{\mathbf{b}}, \quad \boldsymbol{\sigma}_2 = \zeta_{23}^b\, Q_3\, \hat{\mathbf{b}}, \quad \boldsymbol{\sigma}_3 = -\big(\zeta_{13}^b Q_1 + \zeta_{23}^b Q_2\big)\hat{\mathbf{b}}, \qquad \zeta_{ab}^b = -\zeta_{ba}^b, \tag{B.49}$$

so $\boldsymbol{\sigma}\boldsymbol{\sigma}^T = \big[(\zeta_{13}^b Q_3)^2 + (\zeta_{23}^b Q_3)^2 + (\zeta_{13}^b Q_1 + \zeta_{23}^b Q_2)^2\big]\hat{\mathbf{b}}\hat{\mathbf{b}}^T$ corrects only the bb element of the inertia,

$$\mathbf{I}'(Q) = \text{diag}\big(I_a(Q),\, I_b(Q) - s(Q),\, I_c(Q)\big), \qquad s(Q) \equiv (\zeta_{13}^b Q_3)^2 + (\zeta_{23}^b Q_3)^2 + (\zeta_{13}^b Q_1 + \zeta_{23}^b Q_2)^2. \tag{B.50}$$

Note $s(Q) = \mathcal{O}(Q^2)$, so it does not affect the linear-order derivatives (B.48) but does enter the pseudopotential and the exact Jacobian.

Determinant, volume element, and its explicit derivatives. With $\mathbf{I}'$ diagonal in (B.50),

$$\gamma(Q) = \det \mathbf{I}' = I_a(Q)\,\big(I_b(Q) - s(Q)\big)\, I_c(Q), \qquad \sqrt{g} = \sin\theta\, \sqrt{I_a\,(I_b - s)\, I_c}, \tag{B.51}$$

and the vibrational Jacobian carries the explicit derivatives

$$\partial_{Q_b} \ln \gamma = \frac{a_a^b}{I_a} + \frac{a_b^b - \partial_{Q_b} s}{I_b - s} + \frac{a_c^b}{I_c}, \qquad \partial_{Q_3} s = 2\big[(\zeta_{13}^b)^2 + (\zeta_{23}^b)^2\big] Q_3, \;\; \partial_{Q_1} s = 2\zeta_{13}^b\big(\zeta_{13}^b Q_1 + \zeta_{23}^b Q_2\big), \text{ etc.}, \tag{B.52}$$

which are precisely the first-order remainders exhibited under (4.31) and removed by the $\gamma^{1/4}$ rescaling of § 4.9.

Pseudopotential. Since $\mathbf{I}'$ is diagonal, $\boldsymbol{\mu} = \mathrm{diag}\big(1/I_a,\, 1/(I_b - s),\, 1/I_c\big)$ and the trace formula (4.38) is fully explicit:

$$\boxed{\hat{U}(Q) = -\frac{\hbar^2}{8}\Big[\frac{1}{I_a(Q)} + \frac{1}{I_b(Q) - s(Q)} + \frac{1}{I_c(Q)}\Big].} \tag{B.53}$$

Expanding to the order at which it first varies with the coordinates, and using (B.47),

$$\hat{U}(Q) = -\frac{\hbar^2}{8}\Big(\frac{1}{I_a^e} + \frac{1}{I_b^e} + \frac{1}{I_c^e}\Big) + \frac{\hbar^2}{8}\sum_{b=1,2}\Big(\frac{a_a^b}{(I_a^e)^2} + \frac{a_b^b}{(I_b^e)^2} + \frac{a_c^b}{(I_c^e)^2}\Big)Q_b + \mathcal{O}(Q^2). \tag{B.54}$$

The constant is an inconsequential shift of the electronic–vibrational zero; the linear term is a genuine $\hbar^2$ force on the normal coordinates — the physical content of the Watson pseudopotential for this molecule — produced entirely by differentiating $\sqrt{\gamma(Q)}$ through (B.52). Every step from the metric (4.15) to the surface (B.54) used only $\partial_\phi, \partial_\theta, \partial_\chi$ and ∂_{Q_b}.

B.8 Summary

1. The quantized kernel (B.1) expands into the nine products (B.6). Blocks built from mutually commuting, individually Hermitian factors — (T1) $\hat{J}\mu\hat{J}$, (T2) $\hat{L}\mu\hat{L}$, (T4) $\hat{J}\mu\hat{L}$ — are Hermitian in a single ordering, by the adjoint-reversal argument (B.7) alone.

2. Blocks pairing $\hat{\pi}$ with $\hat{J}$ or $\hat{L}$ — (T5), (T6) — are Hermitian only as the symmetric sum $\hat{X} + \hat{X}^\dagger$, which is exactly the ordering the Laplace–Beltrami operator produces by itself [(4.30), (4.35)]. (T3) $\hat{\pi}\mu\hat{\pi}$ is the fortunate middle case, Hermitian because $\hat{\pi}$ is [(B.5)], and equal to the left-ordered $\frac{1}{2}\hat{P}G\hat{P}$ by the sum rule (B.10).

3. No kernel ordering produces a pure multiplicative term. The Watson pseudopotential comes solely from the measure: U_{vib} [(4.34)] plus U_{Cor} [(B.18)] collapse to $-\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$ [(4.38)].

4. Every Hermiticity statement above used only integration by parts, the product rule, and the divergence identities (4.18), (4.26), (B.4) — never a commutator of body-fixed angular momenta.

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